



Part One

DEFINING THE STRUCTURE OF CULTURE

To understand how culture works we need to differentiate two perspectives. The most obvious and immediate impulse is to look for culture *content*. What is the culture about, what are the key values that we need to understand, what are the rules of behavior? Different people have different biases and assumptions about what is important. In the current national context we see a great emphasis on the cultural content pertaining to the role of government, leadership, and management in deciding what is good for everyone and focusing on the values of individual freedom and autonomy. Another culture analysis might, however, say that this is totally irrelevant to what the values are around saving the planet and becoming environmentally responsible. A third person chimes in with the importance of family values and the threat to “our culture” of allowing civil marriage. Parents lament or praise the new values that their children are bringing into the culture, or are just plain puzzled about what this new “millennial” generation is all about. We have to watch our language lest we say something “politically incorrect” about racial or gender issues.

The point is that culture *content*, the values we care about are all over the map. To make some sense of this variety, we have to look first

at the *structure* of culture and develop a perspective on how to analyze the complex cultural landscape we encounter. In the next four chapters I will develop a “model” of the structure of culture. We will analyze several organizational cultures, illustrate how nested they are in larger cultural units. Chapter 1 gives a dynamic definition of culture. Chapter 2 describes the basic three-level model of the “structure” of culture that will be used throughout the rest of the book. In Chapter 3 this model is illustrated with Digital Equipment Corporation, a U.S. computer company that I encountered in its early growth period and in which I could, therefore, observe the growth and evolution of a culture. In Chapter 4 I describe Ciba-Geigy, an old Swiss-German chemical company, that illustrates some of the problems of a mature industry in a very different technology and the impact of national culture. In Chapter 5 I describe the Singapore Economic Development Board, which illustrates both a fusion of Western and Asian national cultures and an organization in the public sector. The cases are intended to highlight that cultures are learned *patterns of beliefs, values, assumptions, and behavioral norms that manifest themselves at different levels of observability*.

HOW TO DEFINE CULTURE IN GENERAL

The Problem of Defining Culture Clearly

Culture has been studied for a long time by anthropologists and sociologists, resulting in many models and definitions of culture. Some of the ways that they have conceptualized the essence of culture illustrate the breadth as well as the depth of the concept. Most of the categories that follow refer primarily to *macro* cultures such as nations, occupations, or large organizations but some are also relevant to *micro* or subcultures. As you will see from the pattern of references, many researchers use several of these definitional categories, and they overlap to a considerable degree. Culture as we will see exists at many levels of “observability.” The categories are arranged roughly according to the degree to which you, as an observer, will be able to see and feel those cultural elements when you observe an organization or group.

- **Observed behavioral regularities when people interact:** The language they use along with the regularities in the interaction such as “Thank you” followed by “Don’t mention it,” or “How is your day going so far,” “Just fine.” Observed interaction patterns, customs, and traditions become evident in all groups in a variety of situations (e.g., Goffman, 1959, 1967; Jones, Moore, & Snyder, 1988; Trice & Beyer, 1993; Van Maanen, 1979).
- **Climate:** The feeling that is conveyed in a group by the physical layout and the way in which members of the organization interact with each other, with customers, or with other outsiders. Climate is sometimes included as an artifact of culture and is sometimes kept as a separate phenomenon

to be analyzed (e.g., Ashkanasy, Wilderom, & Peterson, 2000; Schneider, 1990; Tagiuri & Litwin, 1968; Ehrhart, Schneider, & Macey, 2014).

- **Formal rituals and celebrations:** The ways in which a group celebrates key events that reflect important values or important “passages” by members such as promotion, completion of important projects, and milestones (Trice & Beyer, 1993; Deal & Kennedy, 1982, 1999).
- **Espoused values:** The articulated, publicly announced principles and values that the group claims to be trying to achieve, such as “product quality,” “price leadership,” or “safety” (e.g., Deal & Kennedy, 1982, 1999). Many companies in Silicon Valley such as Google and Netflix announce their culture in terms of such values in all of their recruiting materials and in books about themselves (Schmidt & Rosenberg, 2014).
- **Formal philosophy:** The broad policies and ideological principles that guide a group’s actions toward stockholders, employees, customers, and other stakeholders such as the highly publicized “HP way” of Hewlett-Packard or, more recently, the explicit statements about culture in Netflix and Google (e.g., Ouchi, 1981; Pascale & Athos, 1981; Packard, 1995; Schmidt & Rosenberg, 2014).
- **Group norms:** The implicit standards and values that evolve in working groups, such as the particular norm of “a fair day’s work for a fair day’s pay” that evolved among workers in the Bank Wiring Room in the classic Hawthorne studies (e.g., Homans, 1950; Kilmann & Saxton, 1983).
- **Rules of the game:** These are the implicit, unwritten rules for getting along in the organization, “the ropes” that a newcomer must learn to become an accepted member, “the way we do things around here” (e.g., Schein, 1968, 1978; Van Maanen, 1976, 1979b; Ritti & Funkhouser, 1987; Deal & Kennedy, 1999).
- **Identity and images of self:** How the organization views itself in terms of “who we are,” “what is our purpose,” and “how we do things” (e.g., Schultz, 1995; Hatch, 1990; Hatch & Schultz, 2004).
- **Embedded skills:** The special competencies displayed by group members in accomplishing certain tasks, the ability to make certain things that get passed on from generation to generation without necessarily

being articulated in writing (e.g., Argyris & Schon, 1978; Cook & Yanow, 1993; Peters & Waterman, 1982; Ang & Van Dyne, 2008).

- **Habits of thinking, mental models, or linguistic paradigms:** The shared cognitive frames that guide the perceptions, thoughts, and language used by the members of a group and are taught to new members in the socialization or “onboarding” process as it is now often called (e.g., Douglas, 1986; Hofstede, 1991, 2001, Hofstede, Hofstede, & Minkov, 2010; Van Maanen, 1979).
- **Shared meanings:** The emergent understandings that are created by group members as they interact with each other where the same words used in different cultures can have very different meanings (e.g., Geertz, 1973; Smircich, 1983; Van Maanen & Barley, 1984; Weick, 1995; Weick & Sutcliffe, 2001; Hatch & Schultz, 2004).
- **“Root metaphors” or integrating symbols:** The ways that groups evolve to characterize themselves, which may or may not be appreciated consciously but become embodied in buildings, office lay-outs, and other material artifacts of the group. This level of the culture reflects the emotional and aesthetic response of members as contrasted with the cognitive or evaluative response (e.g., Gagliardi, 1990; Hatch, 1990; Pondy, Frost, Morgan, & Dandridge, 1983; Schultz, 1995).

I have provided these many ways of defining culture to give you a sense that culture covers pretty much everything that a group has learned as it has evolved. When we look at macro cultures (e.g., nations or occupations) and want to describe their cultures, we need all of these specific concepts to capture their culture. However, in moving toward a usable definition of culture that you can apply to the organizations and groups that you will encounter and that you will want to decipher, we need a more integrative dynamic definition that highlights how culture forms and evolves in organizations, subcultures, and micro systems. The foregoing categories will help to define the content of a given culture, but defining them has to be a more dynamic holistic process.

The reason for such a formal definition at this point is to forewarn you that you will find many groups of various sizes with different shared patterns that must be understood on their own terms. You will see articles about how to change or even create cultures that don’t agree with each

A Dynamic Definition of Culture

The culture of a group can be defined as the accumulated shared learning of that group as it solves its problems of external adaptation and internal integration; which has worked well enough to be considered valid and, therefore, to be taught to new members as the correct way to perceive, think, feel, and behave in relation to those problems.

This accumulated learning is a pattern or system of beliefs, values, and behavioral norms that come to be taken for granted as basic assumptions and eventually drop out of awareness.

other or that don't make sense. This definition is deliberately focused on the general process of how *any culture* is learned and will evolve, but in practice you will have to focus on different elements of that formal definition to make sense of the particular organizational situation you encounter. So let's expand on and explain the importance of each component of that definition in preparation for the more detailed analysis of these elements that occur later on in this book.

Accumulated Shared Learning

The most important element of the definition is to note that culture is a *shared* product of *shared* learning (Edmondson, 2012). If you understand that culture is a shared product of shared learning, you will realize several important corollaries that make culture complex. To fully understand a given group's culture, we will need to know what kind of learning has taken place, over what span of time, and under what kinds of leadership. Deciphering such history is impossible with preliterate culture, nations, and some occupations; however, with contemporary organizations and work groups, it is possible and fruitful to begin culture analysis with historical analysis. I will keep referring to "the group," but I mean this to include organizations of all kinds as well.

If learning is shared, all the group forces of identity formation and cohesion come into play in stabilizing that learning because it comes to define for the group who we are and what is our purpose or "reason to be." The various components of what is learned then become a pattern of beliefs and values that give meaning to the daily activities and work of the group. If the

group is successful in achieving its purpose and is internally well organized, it will come to take these beliefs and values along with the accompanying behavioral norms for granted and will teach them to newcomers as the way to think, feel, and behave. In many ways this can be thought of as the group's sense of identity, which has both an external component of how the organization presents itself to the outside and an internal component of what its inner sense of itself is.

Basic Taken-for-Granted Assumptions—The Cultural DNA

The earliest shared learning provides meaning and stability and becomes, in a sense, *the cultural DNA*: the beliefs, values, and desired behaviors that launched the group and made it successful. This early level of beliefs, values, and desired behavior becomes nonnegotiable and turns into taken-for-granted basic assumptions that subsequently drop out of awareness. Such assumptions come to be very stable, serving as the source of later ways of doing things and elaborating the culture. What needs to be mentioned here is that these elements, learned early and composing the cultural DNA, are the source of the group's stability and cannot be changed without changing the group altogether. This point has to be understood at the outset because culture-change programs can work only if they are consistent with the group's cultural DNA.

Solving Problems of External Adaptation and Internal Integration

One of the most consistent findings of the study of groups and organizations is that leaders and members differentiate the “task” of the group from the question of “how we will organize and maintain ourselves as a group?” This arbitrary distinction has taken many forms, such as the “managerial grid,” which separately measures the degree of concern for task and of concern for people, leading to an “ideal” of maximizing both (Blake & Mouton, 1964, 1969; Blake, Mouton, & McCanse, 1989). In extensive studies of problem-solving groups, it was discovered that two kinds of leadership evolved and were necessary for long-range group performance: a task leader and a social-emotional leader who were usually different people within the group (Bales, 1958).

Studies of effective organizations have always shown that successful performance and effective learning hinge on not separating these two dimensions, thinking instead in terms of “socio-technical systems,” in which the external and internal are at least aligned if not integrated. In business organizations, this issue has shown up in concern for a “scorecard” or a “double bottom line” that emphasizes the need for paying attention to both the economic health of the organization and the internal organizational health that allows it to function and maintain itself (Kaplan & Norton, 1992).

One of the great dangers inherent in culture-change programs is to assume that strategy and the external adaptation issues are somehow separate from culture and to focus the desired culture changes just on the *internal* mechanisms by which a group makes life pleasant for itself. All the emphasis recently on analyzing which company to work for creates the risk that you will go to the best company but will be out of a job in a few years because that same company did not understand that its strategy was also part of its culture and failed to evolve that strategy according to the changing needs of the situation (Friedman, 2014).

Solutions That Have Worked Well Enough to Be Considered Valid

Groups are created for a purpose. We huddle together for safety or security or to get something done, and the group’s survival depends on the degree to which it accomplishes its purpose. Groups do not exist in isolation. To get something done requires some kind of action in the various environments in which the group is embedded. As the group acts, it gets feedback on whether or not it is accomplishing its purpose. If it succeeds and continues to succeed, the beliefs, values, and behavior patterns that launched the group will become taken for granted as the way to continue. With age and continued success, those beliefs and values will become part of the identity of the group and will automatically be taught to newcomers as “this is who we are, this is what we do, and these are our beliefs.” Whereas those values and beliefs might have been debated at the launching of the group, they become nonnegotiable and are treated as “assumptions” that new members are expected to adopt as the price of admission to the group.

Perception, Thought, Feeling, and Behavior

As a group grows, has success, and develops an identity, the shared learning process broadens from just the minimum behavior we need to agree on to get the job done to a language, a way to think, and a way to feel. When a company is founded, there will be a common interest focused on the technology, the product or service, and the occupational competencies required to perform. This means that some common ways of thinking and perceiving are present at the outset by the common decision to be a group and do something together.

With success and further shared experience the group develops its own “jargon,” often expressed as shorthand and acronyms, forms of humor, and expressions that symbolize some of the essence of the shared experience. In Digital Equipment Corporation (DEC), a company that we will be referring to frequently, the phrase “Do the right thing” symbolized the value of technical honesty, openness, and really solving the customer’s problems. In Apple the phrase was “Do your own thing,” which meant feel free to contribute in the best way you can but express yourself personally, which, at the time, meant “decorate your office any way you want, bring your pet to work, but do the job well.”

We tend to think of culture as mostly behavioral (i.e., “This is how we do things around here”) and forget that with time and shared learning we come to share how we talk, what we perceive in our relevant environment, how we think about it, and what makes us feel good or bad. The longer the organization has existed, the more the thoughts and emotions of the members come to be alike. This process is most visible at the national level, where we find that subsidiaries of companies that move to new countries have great difficulty in functioning efficiently because of differences in language, thought, and emotional processes. In some companies the corporate culture is so strong and well embedded that the local offices in different countries look like and function exactly the same way as the headquarters organization.

I was once asked to describe the culture of the Swiss-German company Ciba-Geigy to the U.S. subsidiary in New Jersey. I had studied this culture in Basel and gave my Basel speech in New Jersey, which elicited the shocked response: “My God, you have just described us perfectly!”

What You Imply When You Use the Word *Culture*

The concept of culture implies structural stability, depth, breadth, and patterning or integration that results from the fact that culture is for the group a learned phenomenon just as personality and character are for individuals learned phenomena.

Structural Stability. Culture implies some level of structural stability in the group. When we say that something is “cultural,” we imply that it is not only shared but is also stable because it defines the group. I have referred to this as “basic assumptions” and cultural DNA. After we achieve a sense of group identity, which is a key component of culture, it is our major stabilizing force and will not be given up easily. Culture is something that survives even when some members of the organization depart. Cultural DNA is hard to change because group members value stability because it provides meaning and predictability.

At the same time, the more surface elements of culture are defined by the interaction among the group members. The more ritualized of those interactions support the DNA and provide additional stability, but as new conditions arise and as new members with different beliefs, values, and norms enter the group, there will inevitably be both reinforcement and change as new solutions are invented for the problems of internal and external survival. Culture is both stable and dynamic, just as our body is stable if we think of the skeleton and skin and organs but constantly changing if we think of cells and the various bodily processes. The stable parts like our bones can change but not easily or rapidly unless extreme circumstances cause “breaks.” When companies go bankrupt or are taken over by a turnaround manager, the cultural DNA can be destroyed and a new organization can be launched.

Depth. The basic assumptions of a culture are the deepest, often unconscious part of a group and are, therefore, less tangible and less visible. From this point of view, many of the definitions of culture that I reviewed focus too much on the visible manifestations of culture, but they are not the “essence” of what we mean by culture. This essence, best thought of as the cultural DNA, consists of the taken-for-granted, nonnegotiable beliefs, values, and behavioral assumptions. When something is more deeply embedded, that also lends stability.

Breadth. A third characteristic of culture is that after it has developed, it covers all of a group's functioning. Culture is pervasive and influences all aspects of how an organization deals with its primary purpose, its various environments, and its internal operations. As we have pointed out previously, the most common mistake is to limit the concept to the internal workings of the group while forgetting that culture also covers mission, strategy, structure, and basic operational processes. All of these have been the product of shared learning and will limit the kinds of changes the organization can make.

Patterning or Integration. The fourth characteristic that is implied by the concept of culture and that further lends stability is patterning or integration of the elements into a larger paradigm or “gestalt” that ties together the various elements at a deeper level. Culture implies that rituals, values, and behaviors are tied together into a coherent whole, and this pattern or integration is the essence of what we mean by “culture.” Such patterning or integration ultimately derives from the human need to make our environment as sensible and orderly as we can (Weick, 1995). Because disorder or senselessness makes us anxious, we will work hard to reduce that anxiety by developing a more consistent and predictable view of how things are and how they should be. “Organizational cultures, like other cultures, develop as groups of people struggle to make sense of and cope with their worlds” (Trice & Beyer, 1993, p. 4).

However, we will also discover that within the cultural DNA one finds conflicting themes based on different things learned at different times and in different ways. Furthermore, as organizations evolve and develop subgroups, those subgroups develop their own subcultures, which may conflict with each other or with the larger “corporate culture.” As we will see, cultural dynamics can become very complicated.

Taught to New Members: The Process of Socialization or Acculturation

After a group has developed a culture, it will pass elements of this culture on to new generations of group members (Louis, 1980; Schein, 1968; Van Maanen, 1976; Van Maanen & Schein, 1979). Studying what new

members of groups are taught is, in fact, a good way to discover some of the elements of a culture, but we learn about surface aspects of the culture only by this means. This is especially so because much of what is at the heart of a culture will not be revealed in the rules of behavior taught to newcomers. It will be revealed to members only as they gain permanent status and are allowed into the inner circles of the group, where group secrets then are shared.

However, the way people learn and the socialization processes to which they are subjected may indeed reveal deeper assumptions. To reach those deeper levels, we must try to understand the perceptions and feelings that arise in critical situations, and we must observe and interview regular members or “old timers” to get an accurate sense of the deeper-level assumptions that are shared.

Can culture be learned through anticipatory socialization or self-socialization? Can new members discover for themselves what the basic assumptions are? Yes and no. We certainly know that one of the major activities of any new member when she or he enters a new group is to try to decipher the operating norms and assumptions. But this deciphering will be successful only through experiencing the rewards and punishments that are meted out by long-standing members to new members as they experiment with different kinds of behavior. In this sense, there is always a teaching process going on, even though it may be quite implicit and unsystematic.

If the group has not evolved to the point of having shared assumptions, as will sometimes be the case, the new members’ interaction with old members will be a more creative process of building a culture. But once shared assumptions exist, the culture survives through teaching those assumptions to newcomers. In this regard, culture is a mechanism of social control and can be the basis of explicitly manipulating members into perceiving, thinking, and feeling in certain ways (Van Maanen & Kunda, 1989; Kunda, 1992, 2006). Whether or not we approve of this as a mechanism of social control is a separate question that will be addressed later.

Can Culture Be Inferred from Behavior Alone?

Note that the definition of culture that I have given does not include overt behavior patterns, though some such behavior, especially formal rituals,

would reflect cultural assumptions. Instead, this definition emphasizes that the shared assumptions deal with how we perceive, think about, and feel about things. We cannot rely on overt behavior alone, because it is always determined both by the cultural predisposition (i.e., the shared perceptions, thoughts, and feelings that are patterned) and by the situational contingencies that arise from the immediate external environment.

Behavioral regularities can occur for reasons other than culture. For example, if we observe that all members of a group cower in the presence of a large and loud leader, this could be based on biological-reflex reactions to sound and size, individual learning, or shared learning. Such a behavioral regularity should not, therefore, be the basis for defining culture, though we might later discover that, in a given group's experience, cowering is indeed a result of shared learning and therefore a manifestation of deeper shared assumptions. Or, to put it another way, when we observe behavioral regularities, we do not know whether or not we are dealing with a cultural manifestation. Only after we have discovered the deeper layers that I am defining as the essence or DNA of culture can we specify what is and what is not an "artifact" that reflects the culture.

Do Occupations Have Cultures?

The definition provided previously does not specify the size or location of the social unit to which it can legitimately be applied. We know that nations, ethnic groups, religions, and other kinds of social units have cultures. I call these *macro* cultures. Our experience with large organizations also tells us that even globally dispersed corporations such as IBM or Unilever have corporate cultures in spite of the obvious presence of many diverse subcultures within the larger organization.

But it is not clear whether it makes sense to say that medicine or law or accounting or engineering has cultures. If culture is a product of joint learning leading to shared assumptions about how to perform and relate internally, we can see clearly that many occupations do evolve cultures. If there is strong socialization during the education and training period and if the beliefs and values learned during this time remain stable as taken-for-granted assumptions even though the person may not be in a group of occupational peers, then clearly those occupations have cultures. For

most of the occupations that will concern us, these cultures are global to the extent that members are trained in the same way to the same skill set and values. However, we will find that the macro cultures, the nations and religions in which members of those occupations practice, also influence how the occupations are defined—that is, how engineering or medicine is practiced in a particular country. These variations make it that much more difficult to decipher in a hospital, for example, what is national, ethnic, occupational, or organizational.

Where Does Leadership Come In?

Leadership is the key to learning. Learning occurs when something expected is not happening and the individual or the group feels hungry, hurt, disappointed, or in some other way “disconfirmed.” If we are talking about culture formation, learning occurs through the leadership of a founder or entrepreneur who uses his or her personal power to demand some new behavior directed toward achieving some purpose. If the group gets into difficulty, it will again be leadership that will propose something new to try to get out of the difficulty. If the group is successful, the culture will define what is expected of its formal leaders. If the group then gets into difficulty again, formal leaders or other members of the group will demonstrate or demand some new behavior to solve the problem, which may evolve the culture.

The learning mechanism will vary with the nature of the difficulty. If the group is not doing something that it should be doing, the leader provides it; and if the group succeeds, that behavior is reinforced and is eventually justified with the appropriate beliefs and values. If the group is doing something wrong that produces undesirable results, that behavior is punished by the other cultures in the environment and the group learns never to do that again. But again, the learning of something new or stopping something inappropriate will be mediated by leadership behavior. This will be explored further in the subsequent chapters.

Summary and Conclusions

To summarize, the most useful way to arrive at a definition of something as abstract as culture is to think in dynamic evolutionary terms, to think of culture as what the group has learned in its efforts to survive, grow, deal

with its external environment, and organize itself. If we can understand where culture comes from and how it evolves, we can grasp something that is abstract, that exists in a group's unconscious, yet that has a powerful influence on a group's behavior.

Any social unit that has some kind of shared history will have gone through such a learning process and will have evolved a culture. The strength of that culture depends on the length of time, the stability of membership of the group, and the emotional intensity of the actual historical learning experiences they have shared. As we will see in the case examples, leadership is involved in the creation of the culture and at every stage of the organization's growth and maturity.

Suggestions for Readers

- If you are a scholar or researcher, before you plunge into your research consider that you are about to study a complex, patterned, multifaceted human socio-technical system and decide what it is you are really trying to find out, what research method you will use, and how that research method might affect the system.
- If you are a student or potential employee, ask the recruiter about the history of the company and ask to meet some “old timers” to get their sense of how the company came to be.
- If you are a change leader, ask yourself the following question: If the group or organization I am trying to change has a learning history, what can I learn about that history before I begin to plan changes?
- If you are a consultant or helper who has been asked to build or change culture, be sure to ask the potential client what he or she has in mind and get as concrete a picture as possible of what problems the client is trying to solve before you agree to anything.

2

THE STRUCTURE OF CULTURE

Culture in general can be analyzed at several different levels, with the term “level” meaning the degree to which the cultural phenomenon is visible to you as participant or observer. These levels range from the very tangible, overt manifestations that you can see and feel to the deeply embedded, unconscious, basic assumptions that we are defining as the essence of culture or its DNA. In between these layers are various espoused beliefs, values, norms, and rules of behavior that members of the culture use as a way of depicting the culture to themselves and others. The three major levels of cultural analysis are shown in Figure 2.1.

Three Levels of Analysis

Artifacts—Visible and Feelable Phenomena

We think of artifacts as the phenomena that you would see, hear, and feel when you encounter a new group with an unfamiliar culture. Artifacts include the visible products of the group, such as the architecture of its physical environment; its language; its technology and products; its artistic creations; its style, as embodied in clothing, manners of address, and emotional displays; its myths and stories told about the organization; its published lists of values; and its observable rituals and ceremonies.

Among these artifacts is the “climate” of the group. Some culture analysts see climate as the equivalent to culture, but it is better thought of as the product of some of the underlying assumptions and is, therefore, a manifestation of the culture. Observed behavior routines and rituals are also artifacts, as are the organizational processes by which such behavior is made routine. Structural elements such as charters, formal descriptions of how the organization works, and organization charts also belong to the artifact level.

Figure 2.1 The Three Levels of Culture**1. Artifacts**

- Visible and feelable structures and processes
- Observed behavior
 - Difficult to decipher

2. Espoused Beliefs and Values

- Ideals, goals, values, aspirations
- Ideologies
- Rationalizations
 - May or may not be congruent with behavior and other artifacts

3. Basic Underlying Assumptions

- Unconscious, taken-for-granted beliefs and values
 - Determine behavior, perception, thought, and feeling

The most important point to be made about this level of the culture is that it is both easy to observe and very difficult to decipher. The Egyptians and the Mayans both built highly visible pyramids, but the meaning of pyramids in each culture was very different—tombs in one, temples as well as tombs in the other. In other words, observers can describe what they see and feel but cannot reconstruct from that alone what those things mean to the given group. If you are entering a new culture, you will observe lots of things that may or may not make sense to you, and you will not have the insight to figure them out without asking insiders some questions.

It is especially dangerous to try to infer the deeper assumptions from artifacts alone, because your interpretations will inevitably be projections of your own cultural background. For example, when you see a very informal, loose organization, you may interpret that as “inefficient” if your own background is based on the assumption that informality means playing around and not working. Alternatively, if you see a very formal organization, you may interpret that to be a sign of “lack of innovative capacity,” if your own experience is based on the assumption that formality means bureaucracy and standardization.

If you live in the group long enough, the meanings of artifacts gradually become clear and people explain to you “why we do it that way.” If, however, you want to achieve this level of understanding more quickly, you

must ask insiders why they do what they do? You will then get what we are calling the espoused beliefs and values.

Espoused Beliefs and Values

All group learning ultimately reflects someone's original beliefs and values—his or her sense of what ought to be, as distinct from what is. When a group is first created or when it faces a new task, issue, or problem, the first solution proposed to deal with it reflects some individual's own assumptions about what is right or wrong, what will work or will not work. Those individuals who prevail, who can influence the group to adopt a certain approach to the problem, will later be identified as leaders or founders, but the group does not yet have any shared knowledge as a group because it has not yet taken a common action in reference to whatever it is supposed to do. Whatever is proposed will be perceived only as what the leader wants. Until the group has taken some joint action and together observed the outcome of that action, there is not as yet a shared basis for determining whether what the leader wants will turn out to be valid.

For example, if sales begin to decline in a young business, a manager may say, "We must increase advertising" because of her belief that advertising always increases sales. The group, never having experienced this situation before, will hear that assertion as a statement of that manager's beliefs and values: "She believes that when one is in sales trouble it is a good thing to increase advertising." What the leader initially proposes, therefore, cannot have any status other than a value to be questioned, debated, challenged, and tested. If the manager convinces the group to act on her belief and the solution works, then the perceived value that "advertising is good" gradually becomes transformed, first into a shared value or belief and ultimately into a shared assumption (if actions based on it continue to be successful). If this transformation process occurs, group members will usually forget that originally they were not sure and that the proposed course of action was, at an earlier time, just a proposal to be debated and confronted.

Not all beliefs and values undergo such transformation. First of all, the solution based on a given value may not work reliably. Only those beliefs and values that can be empirically tested and that continue to work reliably in solving the group's problems will become transformed into assumptions.

Second, certain value domains—those dealing with the less controllable elements of the environment or with aesthetic or moral matters—may not be testable at all. In such cases, consensus through social validation is still possible, but it is not automatic. Third, the strategy and goals of the organization may fall into this category of espoused beliefs in that there may be no way of testing them except through consensus, because the link between performance and strategy may be hard to prove.

Social validation means that certain beliefs and values are confirmed only by the shared social experience of a group. For example, any given culture cannot prove that its religion and moral system are superior to another culture's religion and moral system, but if the members reinforce each others' beliefs and values, they come to be taken for granted. Those who fail to accept such beliefs and values run the risk of "excommunication," of being thrown out of the group. The test of whether they work or not is how comfortable and anxiety-free members are when they abide by them. In these realms, the group learns that certain beliefs and values, as initially promulgated by prophets, founders, and leaders, "work" in the sense of reducing uncertainty in critical areas of the group's functioning. Moreover, as they continue to provide meaning and comfort to group members, they also become transformed into non-discussible assumptions even though they may not be correlated with actual performance.

The espoused beliefs and moral or ethical rules remain conscious and are explicitly articulated because they serve the normative or moral function of guiding members of the group as to how to deal with certain key situations as well as in training new members how to behave. Such beliefs and values often become embodied in an ideology or organizational philosophy, which then serves as a guide to dealing with the uncertainty of intrinsically uncontrollable or difficult events.

If the beliefs and values that provide meaning and comfort to the group are not congruent with the beliefs and values that correlate with effective performance, we will observe in many organizations espoused values that reflect the *desired* behavior but are not reflected in *observed* behavior (Argyris & Schon, 1978, 1996). For example, a company's ideology may say that it values people and that it has high quality standards for its products, but its actual record in that regard may contradict what it says. In U.S. organizations, it is common to espouse teamwork while actually rewarding

individual competitiveness. Hewlett-Packard's highly touted "The HP way" (Packard, 1995) espoused consensus management and teamwork, but in its computer division, engineers discovered that to get ahead they had to be competitive and political.

So in analyzing espoused beliefs and values, you must discriminate carefully among those that are congruent with the underlying assumptions that guide performance, those that are part of the ideology or philosophy of the organization, and those that are rationalizations or only aspirations for the future. Often espoused beliefs and values are so abstract that they can be mutually contradictory, as when a company claims to be equally concerned about stockholders, employees, and customers, or when it claims both highest quality and lowest cost. Espoused beliefs and values often leave large areas of behavior unexplained, leaving us with a feeling that we understand a piece of the culture but still do not have the entire culture in hand. To get at that deeper level of understanding, to decipher the pattern, and to predict future behavior correctly, we have to understand more fully the category of basic assumptions.

Taken-for-Granted Underlying Basic Assumptions

When a solution to a problem works repeatedly, it comes to be taken for granted. What was once a hypothesis, supported only by a hunch or a value, gradually comes to be treated as a reality. We come to believe that nature really works this way. Basic assumptions, in this sense, are different from what some anthropologists have called "dominant value orientations," in that such dominant orientations reflect the preferred solution among several basic alternatives, but all the alternatives are still visible in the culture, and any given member of the culture could, from time to time, behave according to variant as well as dominant orientations (Kluckhohn & Strodtbeck, 1961). In the United States, the preferred solution is clearly individualism, but teamwork as a means to an end is accepted.

Basic assumptions, in the sense defined here, have become so taken for granted that you find little variation within a social unit. This degree of consensus results from repeated success in implementing certain beliefs and values, as previously described. In fact, if a basic assumption comes to be strongly held in a group, members will find behavior based on any other

premise inconceivable. For example, in a group whose basic assumption is that the individual's rights supersede those of the group, members find it inconceivable to commit suicide or in some other way to sacrifice themselves to the group even if they had dishonored the group. In a capitalist country, it is inconceivable that someone might design a business organization to operate consistently at a financial loss or that it does not matter whether or not a product works.

In an occupation such as engineering, it is inconceivable to deliberately design something that is unsafe; it is a taken-for-granted assumption that things should be safe. Basic assumptions, in this sense, are similar to what Argyris and Schon (1996) identified as “theories-in-use”—the implicit assumptions that actually guide behavior, that tell group members how to perceive, think about, and feel about things. Basic assumptions, like theories-in-use, are generally non-confrontable and non-debatable and hence are extremely difficult to change. To learn something new in this realm requires us to resurrect, reexamine, and possibly change some of the more stable portions of our cognitive structure, a process that Argyris and others have called “double-loop learning,” or “frame breaking” (Argyris & Schon, 1974, 1996).

Such learning is intrinsically difficult because the reexamination of basic assumptions temporarily destabilizes our cognitive and interpersonal world, releasing large quantities of basic anxiety. Rather than tolerating such anxiety levels, we tend to want to perceive the events around us as congruent with our assumptions, even if that means distorting, denying, projecting, or in other ways falsifying to ourselves what may be going on around us. It is in this psychological process that culture has its ultimate power.

Culture as a set of basic assumptions defines for us what to pay attention to, what things mean, how to react emotionally to what is going on, and what actions to take in various kinds of situations. After we have developed and integrated a set of such assumptions, we will have created a “thought world” or “mental map.” We will then be most comfortable with others who share the same set of assumptions and very uncomfortable and vulnerable in situations where different assumptions operate because either we will not understand what is going on, or, worse, we will misperceive and misinterpret the actions of others (Douglas, 1986; Bushe, 2009).

Culture at this level provides its members with a basic sense of identity and defines the values that provide self-esteem (Hatch & Schultz, 2004). Cultures tell their members who they are, how to behave toward each other, and how to feel good about themselves. Recognizing these critical functions makes us aware why “changing” culture is so anxiety provoking.

To illustrate how unconscious assumptions can distort data, consider the following example. If we assume, on the basis of past experience or education, that other people will take advantage of us whenever they have an opportunity, we expect to be taken advantage of, and we then interpret the behavior of others in a way that coincides with those expectations. If we assume that it is human nature to be basically lazy, and if we observe people sitting in a seemingly idle posture at their desk, we will interpret their behavior as “loafing” rather than “thinking out an important problem.” We will perceive absence from work as “shirking” rather than “doing work at home.”

If this is not only a personal assumption but also one that is shared and thus part of the culture of an organization, we will discuss with others what to do about our “lazy” workforce and institute tight controls to ensure that people are at their desks and busy. If employees suggest that they do some of their work at home, we will be uncomfortable and probably deny the request because we will figure that at home they would loaf (Bailyn, 1992; Perin, 1991).

In contrast, if we assume that everyone is highly motivated and competent, we will act in accordance with that assumption by encouraging people to work at their own pace and in their own way. If we see people sitting quietly at their desks, we will assume that they are thinking or planning. If someone is discovered to be unproductive in such an organization, we will make the assumption that there is a mismatch between the person and the job assignment, not that the person is lazy or incompetent. If employees want to work at home, we will perceive that as evidence of their wanting to be productive.

In both cases, there is the potential for distortion, in that the cynical manager will not perceive how highly motivated some of the subordinates really are, and the idealistic manager will not perceive that there are subordinates who are lazy and are taking advantage of the situation. As McGregor (1960) noted many decades ago, such assumptions about “human nature”

become the basis of management and control systems that perpetuate themselves because if people are treated consistently in terms of certain basic assumptions, they come eventually to behave according to those assumptions to make their world stable and predictable.

Unconscious assumptions sometimes lead to ridiculously tragic situations, as illustrated by a common problem experienced by U.S. supervisors in some Asian countries. A manager who comes from a U.S. pragmatic tradition assumes and takes it for granted that solving a problem always has the highest priority. When that manager encounters a subordinate who comes from a cultural tradition in which good relationships and protecting the superior's "face" are assumed to have top priority, the following scenario has often resulted.

The manager proposes a solution to a given problem. The subordinate knows that the solution will not work, but his unconscious assumption requires that he remain silent because to tell the boss that the proposed solution is wrong is a threat to the boss's face. It would not even occur to the subordinate to do anything other than remain silent or, if the boss were to inquire what the subordinate thought, he might even reassure the boss to go ahead and take the action rather than challenge the boss.

The action is taken, the results are negative, and the boss, somewhat surprised and puzzled, asks the subordinate what he would have done or would he have done something different. This question puts the subordinate into an impossible double bind because the answer itself is a threat to the boss's face. He cannot possibly explain his behavior without committing the very sin he was trying to avoid in the first place—namely, embarrassing the boss. He may even lie at this point and argue that what the boss did was right and only "bad luck" or uncontrollable circumstances prevented it from succeeding.

From the point of view of the subordinate, the boss's behavior is incomprehensible because to ask the subordinate what he would have done shows lack of self-pride, possibly causing the subordinate to lose respect for that boss. To the boss, the subordinate's behavior is equally incomprehensible. He cannot develop any sensible explanation of his subordinate's behavior that is not cynically colored by the assumption that the subordinate at some level just does not care about effective performance and therefore must be gotten rid of. It never occurs to the boss that another assumption—such as "you never embarrass a superior"—is operating, and that, to the

subordinate, that assumption is even more powerful than “you have to get the job done.”

If assumptions such as these operate only in an individual and represent his or her idiosyncratic experience, they can be corrected more easily because the person will detect that he or she is alone in holding a given assumption. The power of culture comes about through the fact that the assumptions are shared and, therefore, mutually reinforced. In these instances, probably only a third party or some cross-cultural experiences could help to find common ground whereby both parties could bring their implicit assumptions to the surface. Even after they have surfaced, such assumptions would still operate, forcing the boss and the subordinate to invent a whole new communication mechanism that would permit each to remain congruent with his or her culture—for example, agreeing that, before any decision is made and before the boss has stuck his or her neck out, the subordinate will be asked for suggestions and for factual data that would not be face threatening. Note that the solution has to keep each cultural assumption intact. We cannot, in these instances, simply declare one or the other cultural assumption “wrong.” We have to find a third assumption to allow them both to retain their integrity.

We have dwelled on this long example to illustrate the potency of implicit, unconscious assumptions and to show that such assumptions often deal with fundamental aspects of life—the nature of time and space; human nature and human activities; the nature of truth and how we discover it; the correct way for the individual and the group to relate to each other; the relative importance of work, family, and self-development; the proper role of men and women; and the nature of the family.

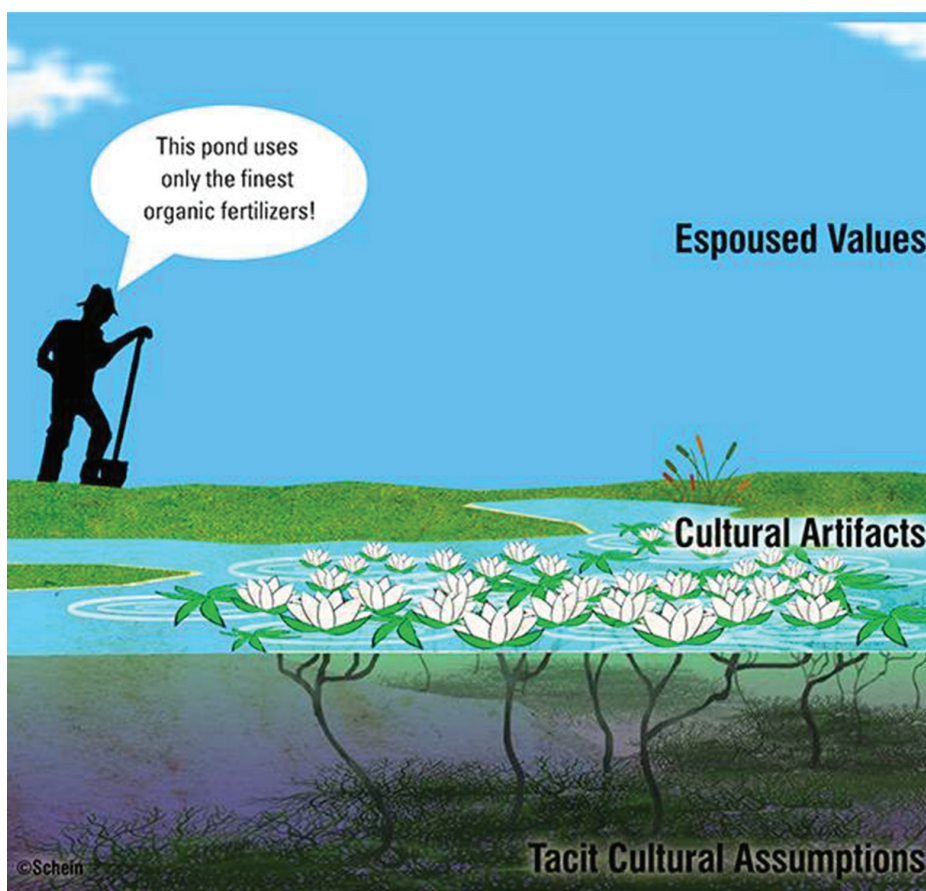
Broader assumptions about human nature often derive from the larger culture in which the organization is embedded or from occupational units that cut across organizations. In the United States, the assumption that meetings are a waste of time derives very much from our pragmatic rugged individualism, which works both against group and team work and immediately types meetings as something to be avoided, even as complex tasks become more interdependent and require more meetings.

The Metaphor of the Lily Pond

We can summarize this three-level model with a metaphoric lily pond. The blossoms and the leaves on the surface of the pond are the “artifacts”

that we can see and evaluate. The farmer who has created the pond (the leadership) announces what he expected and hoped for in the way of leaves and blossoms and will provide publicly accepted beliefs and values to justify the outcome. The farmer may or may not be consciously aware that the outcome is really a result of how the seeds, the root system, the quality of the water in the pond, and the fertilizers he put in combined to create the blossoms and leaves. This lack of awareness of what actually produces the results may not matter if the announced beliefs and values are congruent with how the leaves and blossoms turned out.

Figure 2.2 The Lily Pond as a Metaphor for Levels of Culture



Source: Artwork by Jason Bowes - Human Synergistics

However, if the observer notes a discrepancy between what the farmer claims and what actually comes up as blossoms, they will both have to examine what is present in the water and in the root system. And if they want different color blossoms, painting them a different color will not work; they will have to examine how to change the seeds, the water quality, the fertilizer—that is, the invisible DNA of the pond. Leaders who want to change culture cannot do so by painting the blossoms or pruning the leaves. They have to locate the cultural DNA and change some of that.

Given this structural model one can analyze any culture, or, for that matter, any individual's cultural identity. Let's look briefly at how this would apply at the individual or group micro-system level and then in subsequent chapters apply it to organizations and larger cultural units.

The Individual from a Cultural Perspective

The individual as a cultural entity can be analyzed in terms of artifacts, espoused beliefs and values, and underlying basic assumption. We all carry within us assumptions about the state of the world and about the correct ways to engage in relationships. Some of those assumptions about relationship have come to be taken for granted and fall into the realm of the unconscious because we learned early some of the basic rules of how to get along in different kinds of situations. These assumptions and rules derive from the macro culture in that every society has learned from its own history what level of communication and openness is workable for people to get along.

All societies (i.e., macro cultures) evolve rules of etiquette, good manners, and tact that specify what is or is not appropriate to say in any given situation. Most of us are, therefore, walking repositories of rules that were taught to us when young and that represent early layers of cultural socialization. We learn as part of our acculturation into the family that in the interests of getting along with each other, it is important to withhold some of our perceptions and feelings because to say them out loud might hurt or offend others. And if we hurt others, that permits them to hurt us back, which makes social life generally too dangerous. We learn that some of these things can be said to friends and even more can be said to intimates.

However, the basic assumptions about *why* you cannot say certain things remain below consciousness, and the process by which you learned them is probably totally forgotten.

When we enter into a therapeutic or personal-development program, the leader and the setting usually create a “cultural island” in which some of the societal rules can be suspended and people are encouraged to be more open about what they normally would withhold. When the tasks we are asked to perform in a group require a high degree of collaboration, the team learning process or “teaming” (Edmondson, 2012) similarly creates conditions where some of our basic assumptions have to be surfaced. The best example would be to give team members feedback on how we react to their participation and to own up to our own doubts and fears in relation to task accomplishment. I have called this “here-and-now humility” to indicate that in such team situations formal status and rank become less important than patterns of who is dependent on whom at a given moment in accomplishing a task (Schein, 2016).

In summary, as individuals we can all be observed at the artifact level, we all have our espoused beliefs and values that may or may not be consistent with our behavior, and we all have deeper-level assumptions about why we do what we do. It is the degree of alignment or congruity between the three levels that determine how an individual’s “sincerity,” or “integrity” is judged by others.

The Group or Micro System from a Cultural Perspective

Groups also evolve “hidden agendas,” “have elephants in the room,” and, in various ways, espouse beliefs and principles to justify their overt behavior. If we apply the three-level model to group behavior by analyzing whether or not the observed behavior matches the espoused beliefs and values, we discover discrepancies that reveal the basic assumptions level (Bion, 1959; Marshak, 2006; Kantor, 2012).

A simple but telling example occurred in a company manufacturing team that was dedicated to good team work and espoused a climate of relevant participation by all members. Over several meetings I observed that one member was consistently ignored after he tried to say something, was never called on, and seemed to be very much on the margin. I pointed this

out at one of the meetings and was met with a stony silence, a pause, and then a continuation of the discussion as if nothing had happened.

After the meeting, the chair pointed out to me that this member had been one of the important inventors of several of the company's products, was still too young to be early retired, and was still potentially useful to have around for consultation, but there was no place to "park" him except in this particular group. In early meetings they had welcomed him and jointly agreed that he was welcome to participate but that he would probably find that most of his ideas were now obsolete. He understood and accepted this.

My intervention in calling attention to this embarrassed everyone by surfacing the basic assumption "we accept you as a member but we all understand that you will not be a real contributing member of the group." Any discussion of this assumption would lead to further embarrassment for all concerned. It had become part of the group's culture to accept this person as a member without, however, feeling obligated to take his ideas seriously. The group had evolved the behavioral rule of "you must be polite and pay attention to him but you don't have to use his ideas."

Do all groups have cultures? It depends on the degree to which a given group has a shared history of learning together. A group that has constant change of membership and has not had to learn to do anything together will not have a culture. But any group that has a shared task, more or less constant membership, and some common history of learning together will have its own subculture as well as being nested in the culture of the organizational unit it is in and in the macro cultures of the occupations of its members, the organization, and the nation.

Summary and Conclusions

This chapter presents a three-level model of culture as the way to describe and analyze any cultural phenomenon, whether we are talking about an individual, a micro system, a subculture, an organization, or a macro culture. It is important to differentiate the observed and experienced "artifacts" from the "espoused values" and from the "basic underlying assumptions" that ultimately drive the observed behavior.

Suggestions for Readers

- If you are a scholar or researcher, try to classify all that you observe and know about the group that you are a member of into the basic categories of artifacts, espoused values, and basic assumptions. What additional questions do you need to ask of your colleagues to decipher the basic assumptions?
- If you are a student or potential employee, take a potential organization you are interested in, visit it to gather impressions and feelings, and then see whether what the organization claims fits what you have observed and felt. If you see discrepancies, ask questions to get at the basic assumptions.
- If you are a change leader, bring together a representative group of members of the organization you are trying to change and ask them to identify as many behavioral artifacts of the organization as they can. List these on flip charts. Then ask the group to identify the major espoused values of the organization and compare those values with the artifacts on the charts. Are they consistent? If you find discrepancies, ask the group to identify what the deeper assumption might be that would explain the artifacts, especially observed routine behavior.
- If you are a consultant or helper and are sure you know what specific changes the change leaders have in mind, invite them to bring together a group from their organization and take it through the preceding exercise to determine where identified beliefs, values, and assumptions might aid or hinder the proposed change program.

3

A YOUNG AND GROWING U.S. ENGINEERING ORGANIZATION

How culture works and how to analyze and assess cultural phenomena is best illustrated through cases that represent different stages of organizational evolution. In this chapter I review the case of an organization that I was fortunate enough to encounter in its youth and able to follow through its entire life cycle. At one level this is an “old” case from the 1960s, but the culture dynamics I encountered then continue to be visible in companies that I observed in recent years and seem to characterize technically based start-ups.

Case 1: Digital Equipment Corporation in Maynard, Massachusetts

Digital Equipment Corporation (DEC) was the first major company to introduce interactive computing in the mid-1950s, and it became a very successful manufacturer of what came to be called “mini computers.” It was located primarily in the northeastern part of the United States, with headquarters in an old mill in Maynard, Massachusetts, but it had branches throughout the world. At its peak, it employed more than 100,000 people, with sales of \$14 billion. In the mid-1980s it became the second largest computer manufacturer in the world after IBM. The company ran into major financial difficulties in the 1990s and was eventually sold to the Compaq Corp. in 1998. Compaq was in turn acquired by Hewlett-Packard in 2001.

There were innumerable stories written about why and how DEC “failed,” but few of them provided a cultural perspective on either its rise or its failure. I was involved with DEC as a consultant from 1966 to 1992, and therefore was privy to much of the inside story of how this

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company grew, peaked, and declined (Schein, 2003). I was a consultant to the founder, Ken Olsen, and to the various executives over this entire period, which provided a unique opportunity to see cultural dynamics in action over most of the life of this company. DEC's history is a prime example of how the deeper cultural layers, the basic assumptions, explain both the rise and fall of the company and will be used throughout this book as a major case example that illustrates macro and micro culture interactions.

In this chapter we begin by analyzing the company's culture structurally, using the framework provided in Chapter 2. In later chapters I will refer to various cultural forces that illuminate DEC as a start-up, DEC as a mid-life company, and DEC in decline. Start-ups and old companies have received a lot of attention in the organizational literature, but there are not many studies of an entire life cycle of a single company under its founder. Such studies become especially important as organization theory puts increasing attention on agility and "ambidexterity" as a key property of organizations that have survived for a long time (O'Reilly & Tushman, 2016).

What these authors show with the use of this concept is that long-range survival seems to hinge on the ability to manage both the existing business, which has been the reason for success thus far, and at the same time to develop a new business that is more responsive to changing environmental conditions. If the organization cannot do that for itself, it will inevitably attract competitors who will "disrupt" the present business by creating new businesses that are more adaptive and will, therefore, eventually put the old business into decline (Christensen, 1997). So with this theoretical background, let's meet DEC.

Artifacts: Encountering the Company

To gain entry into any of DEC's many buildings, you had to sign in with a guard who sat behind a counter where there were usually several people chatting, moving in and out, checking the badges of employees who were coming into the building, accepting mail, and answering phone calls. After signing in, you waited in a small, casually furnished lobby until the person you were visiting came personally or sent a secretary to escort you into the work areas.

What I recall most vividly from my first encounters with this organization was the ubiquitous open-office architecture, the extreme informality of dress and manners, a very dynamic environment in the sense of rapid pace, and a high rate of interaction among employees, seemingly reflecting enthusiasm, intensity, energy, and impatience. As I would pass cubicles or conference rooms, I would get the impression of openness. There were very few doors, and I learned later that Ken Olsen, the founder, had forbidden doors on engineers' offices. The company cafeteria spread out into a big open area where people sat at large tables, hopped from one table to another, and obviously were intensely involved in their work even at lunch. I also observed that there were many cubicles with coffee machines and refrigerators in them and that food seemed to be part of most meetings. For morning meetings one or another member brought boxes of fresh donuts for all to enjoy.

The physical layout and patterns of interaction made it very difficult to decipher who had what rank, and I was told that there were no status perquisites such as private dining rooms, special parking places, or offices with special views and the like. The furniture in the lobbies and offices was very inexpensive and functional. The company was mostly headquartered in an old industrial building. The informal clothing worn by most managers and employees reinforced this sense of economy and egalitarianism.

I had been brought into DEC by Ken Olsen "to help the top management team improve communication and group effectiveness." As I began to attend the regular staff meetings of the senior management group, I was quite struck by the high level of interpersonal confrontation, argumentativeness, and conflict. Group members became highly emotional at the drop of a hat, interrupted each other constantly, and seemed to become angry at each other, though it was also noticeable that such anger did not carry over outside the meeting.

With the exception of the president and founder, Ken Olsen, there were very few people who had visible status in terms of how people deferred to them. Olsen himself, through his informal behavior, implied that he did not take his position of power all that seriously. Group members argued as much with him as with each other and even interrupted him from time to time. His status did show up, however, in the occasional lectures he

delivered to the group when he felt that members did not understand something or were “wrong” about something. At such times, Olsen could become very emotionally excited in a way that other members of the group never did. I learned from further observation that this style of running meetings was typical and that meetings were very common, to the point where people would complain about all the time spent in committees. At the same time, they would argue that without these committees they could not get their work done properly.

My own reactions to the company and these meetings also must be considered as artifacts to be documented. It was exciting to be attending top management meetings and surprising to observe so much behavior that seemed to me dysfunctional. The level of confrontation I observed made me quite nervous, and I had a sense of not knowing what this was all about, but it also provided me an agenda as a consultant: how to fix this dysfunctional group in terms of what I knew from my training were the characteristics of effective groups.

The company was organized as a matrix, one of the earliest versions of this type of organization in terms of functional units and product lines, but there was a sense of perpetual reorganization and a search for a structure that would “work better.” Structure was viewed as something to tinker with until you got it right. There were many levels in the technical and managerial hierarchy, but I sensed that the hierarchy was just a convenience, not something to be taken very seriously.

However, the communication structure was taken very seriously. There were many committees already in existence, and new ones were constantly being formed. The company had an extensive email network that functioned worldwide; engineers and managers traveled frequently and were in constant telephone communication with each other. Olsen would get upset if he observed any evidence of under-communication or miscommunication. To make communication and contact easier, DEC had its own “air force” of several planes and helicopters. Ken Olsen was a licensed pilot and flew his own plane to a retreat in Maine for recreation.

Analytical Comment. Many other artifacts from this organization are described subsequently; but for the present, this will suffice to give a flavor

of what I encountered at DEC. The question now is, what does any of it mean? I knew that I reacted very positively to the informality but very negatively to the unruly group behavior, but I did not really understand why these things were happening and what significance they had for members of the company. To gain some understanding, I had to get to the next level: the level of espoused beliefs, values, and behavioral norms.

At this point I thought I was observing subcultures (the various work areas) and micro cultures (the various group meetings) that reflected primarily the technology driving the business—that is, inventing computers that would be interactive, could sit on a desk, and could create a new industry. I was also seeing the personal style of the founder, which seemed to be a reflection of the New England Yankee macro culture. What would I learn if I asked questions?

Espoused Beliefs, Values, and Behavioral Norms

As I talked to people at DEC about my observations, especially those things that puzzled and scared me, I began to elicit some of the espoused beliefs and values by which the company ran. Many of these were embodied in slogans or in parables that Olsen wrote from time to time and circulated throughout the company. For example, a high value was placed on *personal responsibility*. If someone made a proposal to do something and it was approved, that person had a clear obligation to do it or, if it was not possible to do, to come back and renegotiate. The phrase “*He who proposes, does*” was frequently heard around the organization.

Employees at all levels were responsible for thinking about what they were doing and were enjoined at all times to “*do the right thing*,” which, in many instances, meant being insubordinate. If the boss asked you to do something that you considered wrong or stupid, you were supposed to “*push back*” and attempt to change the boss’s mind. If the boss insisted and you still felt that it was not right, you were supposed to not do it and take your chances on your own judgment. If you were wrong, you would get your wrist slapped but would gain respect for having stood up for your own convictions. Because bosses knew these rules, they were, of course, less likely to issue arbitrary orders, more likely to listen to you if you pushed back, and more likely to renegotiate the decision. So actual insubordination

was rarely necessary, but the principle of thinking for oneself and doing the right thing was very strongly reinforced.

It was also a rule that you should not do things without *getting “buy-in”* from others who had to implement the decision, who had to provide needed services, or who would be influenced by it. Employees had to be very individualistic and, at the same time, very willing to be team players; hence, the simultaneous feeling that committees were a big drain on time but were indispensable. To reach a decision and to get buy-in, the individual had to convince others of the validity of his or her idea and be able to defend it against every conceivable argument, which caused the high levels of confrontation and fighting that I observed in groups.

However, after an idea had stood up to this level of debate and survived, it could then be moved forward and implemented, because everyone was now convinced that it was the right thing to do. This took longer to achieve, but led to more consistent and rapid action. If somewhere down the hierarchy the decision “failed to stick” because someone was not convinced that it was “the right thing to do,” that person had to push back, his or her arguments had to be heard, and either he or she had to be convinced or the decision had to be renegotiated up the hierarchy.

In asking people about their jobs, I discovered another strong value: each person should *figure out the essence of your job* and be very clear about it. Asking the boss what was expected was considered a sign of weakness. If your own job definition was out of line with what the group or department required, you would hear about it soon enough. The role of the boss was to set broad targets, but subordinates were expected to take the initiative to figure out how best to achieve them. This value required a lot of discussion and negotiation, which often led to complaints about time wasting; at the same time, everyone defended the value of doing things in this way and continued to defend it even though it created difficulties later in DEC’s life.

I also found out that people could fight bitterly in group meetings, yet remain very good friends. There was a feeling of being a tight-knit group, a kind of extended family under a strong father figure, Ken Olsen, which led to the norm that *fighting does not mean that people dislike or disrespect each other*. This norm seemed to extend even to “bad-mouthing” each other; people would call each other “stupid” behind each others’ backs or say that

someone was a real “turkey” or “jerk,” yet they would respect each other in work situations.

Olsen often criticized people in public, which made them feel embarrassed, but it was explained to me that this only meant that you should work on improving your area of operations, not that you were really in disfavor. In fact, people quipped that it was better to have Ken criticize you than not to notice you. Even if someone fell into disfavor, he or she was viewed merely as being in the “penalty box.” Stories were told of managers or engineers who had been in this kind of disfavor for long periods of time and then rebounded to become heroes in some other context.

When managers talked about their products, they emphasized quality and elegance. The company was founded by engineers and was dominated by an engineering mentality in that the value of a proposed new product was generally judged by whether the engineers themselves liked it and used it, not by external market surveys or test markets. DEC engineers loved sophisticated customers like scientists and lab managers who could relate to the complex products, give good feedback, and thereby stimulate product improvements. Ordinary customers were talked about in a rather disparaging way, especially those who might not be technically sophisticated enough to appreciate the elegance of the product that had been designed.

Olsen emphasized *absolute integrity in designing, manufacturing, and selling*. He viewed the company as highly ethical, and he strongly emphasized the work values associated with the Protestant work ethic—honesty, hard work, high standards of personal morality, professionalism, personal responsibility, integrity, and honesty. Especially important was being honest and truthful in their relations with each other and with customers. As this company grew and matured, it put many of these values into formal statements and taught them to new employees. They viewed their culture as a great asset and felt that the culture itself had to be taught to all new employees (Kunda, 1992, 2006).

Analytical Comments. There is a great temptation at this point to conclude that we now understand the DEC culture. I now “knew” what the espoused values and principles were but did not really understand “why” some of these values were so strongly held. I was also struck by the fact

that those values represented simultaneously the macro culture of academia, in which ideas always have to be attacked and tested; the macro occupational engineering culture, in which elegance is a high value; and the micro culture of a start-up, in which the founder's values and operational methods are the primary influence on how the organization evolves. Ken Olsen was a very puritanical New Englander, and he infused those personal values into the organization. For example, there was no alcohol allowed at off-site meetings. Olsen strongly emphasized frugality, drove a very inexpensive car, and disallowed perquisites like private assigned parking spaces and executive dining rooms.

In figuring out which of these nested values are dominant in determining the evolution of the organization, it is important to note that other technical start-ups such as Hewlett-Packard, Apple, Microsoft, and Google had very similar cultures at the early stages of their development.

Basic Assumptions: The Basic DEC Paradigm

To understand the implications of these values and to show how they relate to overt behavior, we must seek the underlying assumptions and premises on which this organization was based (see Figures 3.1 and 3.2).

Figure 3.1 DEC's Cultural Paradigm: Part One

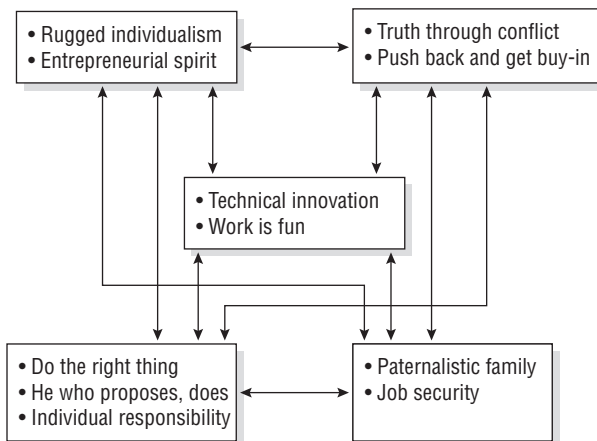
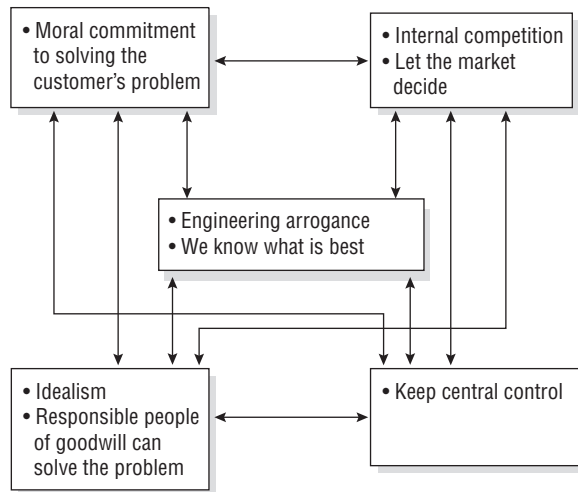


Figure 3.2 DEC's Cultural Paradigm: Part Two

Analytical Comments. Only by grasping these first five assumptions can we understand, for example, why my initial interventions of trying to get the group to be “nicer” to each other in the communication process were politely ignored. I was seeing the group’s “effectiveness” in terms of my values and assumptions of how a “good” group should act. The DEC senior management committee was trying to reach “truth” and make valid decisions in the only way they knew how and by a process that they believed in. The group was merely a means to an end; the real process going on in the group was a basic, deep search for solutions in which they could have confidence because they stood up even after intense debate.

After I shifted my focus to helping them in this search for valid solutions, I figured out what kinds of interventions would be more relevant, and I found that the group accepted them more readily. For example, I began to emphasize agenda setting, using the flip-chart to focus on ideas and to keep them visible in front of the group; time management; clarifying some of the debate; summarizing; consensus testing after the debate was running dry; and a more structured problem-solving process. The interrupting, the emotional conflicts, and the other behaviors I observed initially continued, but the group became more effective in its handling of information and in reaching consensus. It was in this context that I gradually developed the

philosophy of being a “process consultant” instead of trying to be an expert on how groups should work (Schein, 1969, 1988, 1999a, 2003, 2016).

Additional Basic Assumptions

As I learned more about DEC, I also learned that the cultural DNA contained another five key assumptions, shown in Figure 3.2. These five additional assumptions reflected some of the group’s beliefs and values pertaining to customers and marketing:

1. The only valid way to sell a product is to find out what the customer’s problem is and to solve that problem, even if that means selling less or recommending another company’s products.
2. People can and will take responsibility and continue to act responsibly no matter what.
3. The market is the best decision maker if there are several product contenders, which also implies that internal competition in product development is desirable.
4. Even as the company becomes very large and differentiated, it is desirable to keep some central control rather than divisionalizing. I think this was an imposition of Ken Olsen’s personal needs as a founder instead of a strategic decision.
5. DEC engineers “know best” what a good product is, based on whether or not they personally like working with that product.

Analytical Comments. These 10 assumptions can be thought of as the DEC cultural paradigm. What is important in showing the interconnections is the fact that single elements of the paradigm could not explain how this organization was able to function. It was only by seeing the combination of assumptions—around individual creativity, group conflict as the source of truth, individual responsibility, commitment to each other as a family, commitment to innovation and to solving customer problems, and belief in internal competition and central control—that the observable day-to-day behavior could be explained. It is this level of basic assumptions and their

interconnections that defines some of the *essence* of the culture—the key genes of the cultural DNA at this stage of DEC’s development.

This paradigm is a mixed combination of U.S. macro-cultural values of individualism, competition, and pragmatism and family values such as loyalty, frugality, truth, and commitment to customers as represented by Ken Olsen and the engineers he hired. It illustrates the enormous power of the founder of an organization in how he infuses it with the values and assumptions he brings with him from his own cultural background.

How general was this paradigm in DEC? That is, if we were to study various micro systems such as the workers in the plants, salesmen in geographically remote units, engineers in technical enclaves, and so on, would we find the same assumptions operating? One of the interesting aspects of the DEC story is that at least for its first 20 or so years, this paradigm would have been observed in operation across most of its rank levels, functions, and geographies. Much of it was explicitly taught to all newcomers through periodic “boot camps,” and much of it was written down as the DEC culture. Some of these assumptions were modified in functions such as sales and service where the pragmatic needs to relate well to customers created new and different subcultural elements—hierarchy, status rituals, more rapid decision making, and more discipline. As we will see subsequently, as DEC grew, aged, and evolved, some basic elements of the DEC culture began to change, whereas other elements that did not change, even as they became dysfunctional in the changing market environment, led ultimately to DEC’s decline.

Because of my continuing contact with this company I was able to construct the detailed picture of the organization that is presented in the figures. This level of detail was needed to understand some of the puzzling behavior I had observed and to describe the organization to outside observers. But it must be noted that much of the current emphasis in culture research focuses on how one might *change* culture to improve performance. To change culture requires the insider change leaders to understand their culture at a detailed level and especially to identify various stable elements that were the source of the company’s success, but it is not important for the researcher or outside observer to understand the culture in as much detail as I was able to gather. The consultant, the future employee, the investor, the supplier, or the customer needs to know some elements of an

organization's culture, but he or she does not need to invest in the kind of clinical study that I was able to do in this case.

Summary and Conclusions

The crucial insight to take away from this case is that a young company's culture provides identity, meaning, and daily motivation. If the company is successful, that culture will become very strong and explicitly part of its identity. We see today articles and books by many organizations explicitly touting their culture and suggesting that it is the culture that is making those organizations successful. The DEC story should remind us that we cannot really make generalizations about culture without specifying the age, size, and underlying technology of the company, because each of those factors played a role in what the DEC culture became. We also have to consider in what way the given company culture is nested in various occupational and national macro cultures.

A second important lesson is that the presence of the founder or entrepreneur is a strong stabilizing force for the culture. The implication is that we cannot make generalizations about culture without specifying whether we are talking about a first- or a second-generation company still run by the founder or about a company run by general managers who have been appointed by boards and have worked their way up the managerial ladder. The stability of DEC's commitment to innovation even in the face of markets that wanted simpler turnkey products could be explained in part by the presence of the founder, whose deep assumptions and beliefs were difficult to challenge.

In the following chapter we will look at an entirely different kind of organization at a different stage of development and in a different industry.

Suggestions for Readers

For all readers, the main thing to do at this point is to *reflect* on the complexity of this organization's culture and how its micro and subcultures are nested in both the U.S. culture and the culture of engineering.

Can you think of other organizations that resemble DEC? What is similar and what is different, and why?

4

A MATURE SWISS-GERMAN CHEMICAL ORGANIZATION

A mature organization is culturally different in many ways. The main differences result from size, age, and that it is managed entirely by promoted general managers (not founders or their children). When we talk about culture, we have to remind ourselves to specify the kind of organization, the macro cultures it is nested in, and how big and old it is. This particular case is also one from the 1980s, but, as in the case of DEC, it shows in a very mature, large, diversified organization how different the cultural issues and cultural dynamics are, and still how representative they are of contemporary large mature organizations.

Case 2: Ciba-Geigy Company in Basel, Switzerland

The Ciba-Geigy Company in the late 1970s and early 1980s was a Swiss multinational, multidivisional, geographically decentralized chemical company dealing with pharmaceuticals, agricultural chemicals, industrial chemicals, dyestuffs, photographic chemicals, and some technically based consumer products. It eventually merged with a former competitor, Sandoz, to become what is today—Novartis. My consulting initially on career development evolved into numerous other consulting activities that lasted into the mid-1980s and focused on some major culture change.

Artifacts—Encountering Ciba-Geigy

My initial encounter with this company was through a telephone call from its head of management development, Dr. Jürg Leupold, inquiring whether I would be willing to give a talk to its annual meeting in Switzerland. Ciba-Geigy invited their top 40 to 50 worldwide executives along with one

or two outsiders to a three-day meeting held at a Swiss resort. The purpose was to review strategy and operations and to stimulate the group by having outside lecturers present on topics of interest to the company. Dr. Leupold had asked me to give lectures and do some structured exercises around my research on career anchors at the 1979 annual meeting of its top executives (Schein, 1978, Schein & Van Maanen, 2013). The CEO, Dr. Samuel Koechlin, liked the fact that this research had shown that people are in their jobs for different reasons (the career anchors), but that in each kind of job one could be creative.

Koechlin was Swiss, but he had spent part of his career in the company's U.S. subsidiary and had become interested in all the emphasis he had observed in the U.S. culture on creativity and innovation. He requested that I visit him prior to the annual meeting to discuss how best to use the career-anchor exercise and to "test our chemistry with each other." I made a special flight to Basel to spend the night with him and his family. We decided that the career anchor booklet and job or role planning exercises were to be translated into German so that all the participants could do the exercise and discuss their career anchors at the annual meeting. I would lecture and draw out the implications for creativity and innovation following the exercises.

I was "briefed" by further phone conversations with Dr. Leupold whose first name was Jürg, but I never felt it was appropriate to use that. I learned that the company was run by a board of directors and an internal executive committee of nine people who were legally accountable as a group for company decisions. The chairman, Dr. Koechlin, functioned as the CEO, but the committee made most decisions by consensus. Each member of the committee had oversight responsibility for a division, a function, and a geographic area, and these responsibilities rotated every few years. Both Ciba and Geigy had long histories of growth and had merged in 1970. The merger was considered to be a success, but there were still strong identifications with the original companies, according to many managers. The CEO of Novartis when I asked him in 2006 how the Ciba-Geigy/Sandoz merger went said: "That merger is going fine, but I still have Ciba people and Geigy people!"

My first visit to Ciba-Geigy headquarters offered a sharp contrast to what I had encountered at DEC. I was immediately struck by the formality

as symbolized by large gray stone buildings, heavy doors that were always closed, and stiff uniformed guards in the main lobby. This spacious, opulent lobby was the main passageway for employees to enter the inner compound of office buildings and plants. It had high ceilings, large heavy doors, and a few couches in one corner to serve as a waiting area.

Upon entering the Ciba-Geigy lobby, I was asked by the uniformed guard to check in with another guard who sat in a glassed-in office. I had to give my name and state where I was from and whom I was visiting. The guard then asked me to take a seat and to wait until an escort could take me to my appointed place. As I sat and waited, I noticed that the guard seemed to know most of the employees who streamed through the lobby or went to elevators and stairs leading from it. I had the distinct feeling that any stranger would have been spotted immediately and would have been asked to report as I had been.

Dr. Leupold's secretary arrived in due course and took me up the elevator and down a long corridor of closed offices. Each office had a tiny nameplate that could be covered over by a hinged metal plate if the occupant wanted to remain anonymous. Above each office was a light bulb, some of which showed red and some green. I asked on a subsequent visit what this meant; I was told that if the light was out, the person was not in; if it was green, it was okay to knock; and if it was red, the person did not want to be disturbed under any circumstances.

We went around a corner and down another corridor and did not see another soul during the entire time. When we reached Dr. Leupold's office, the secretary knocked discreetly. When he called to come in, she opened the door, ushered me in, then went to her own office and closed the door. I was offered some tea or coffee, which was brought by the secretary on a large formal tray with china accompanied by a small plate of excellent cookies. I mention that they were "excellent" because it turned out that good food was very much part of Ciba-Geigy's presented identity. Whenever I visited offices in later years in Paris and London, I was always taken to three-star restaurants.

Following our meeting, Dr. Leupold took me to the executive dining room in another building, where we again passed guards. This was the equivalent of a first-class restaurant, with a hostess who clearly knew everyone, reserved tables, and provided discreet guidance on the day's specials.

Aperitifs and wine were offered with lunch, and the whole meal took almost two hours. I was told that there was a less fancy dining room in still another building and an employee cafeteria as well, but that this dining room clearly had the best food and was the right place for senior management to conduct business and to bring visitors.

Ciba-Geigy managers came across as very serious, thoughtful, deliberate, well prepared, formal, and concerned about protocol. I learned later that whereas DEC allocated rank and salary fairly strictly to the actual job being performed by the individual employee, Ciba-Geigy had a system of managerial ranks based on length of service, overall performance, and the personal background of the individual rather than on the actual job being performed at a given time. Rank and status therefore had a much more permanent quality in Ciba-Geigy, whereas in DEC, fortunes could rise and fall precipitously and frequently with job assignment.

In Ciba-Geigy meetings, I observed much less direct confrontation and much more respect for individual opinion. Meetings were geared to information transmission rather than problem solving. Recommendations made by managers in their specific area of accountability were generally respected, accepted, and implemented. I never observed insubordination, and I had the impression that it would not be tolerated. Rank and status thus clearly had a higher value in Ciba-Geigy than in DEC, whereas personal negotiating skill and the ability to get things done in an ambiguous social environment had a higher value in DEC.

Analytical Comments. It was striking how different the initial encounters with this organization were from my first contacts with DEC and how hard it was from the beginning to determine whether this was a reflection of the Swiss-German macro culture, the effect of chemical technology (the basis of all their products), or the history of the company (which included the major merger of Ciba and Geigy), or the fact that the current leader had been “Americanized” through his years in the U.S. subsidiary.

I had the impression that things were very tightly organized and carefully planned; still, to meet Koechlin at his home with his family and spend the night contrasted sharply with my DEC experience. In my entire time as a consultant with DEC I never met any members of Ken Olsen’s family nor the families of any other executives. It highlights the fact that even a

concept like “informal” or “formal” can mean very different things in different macro cultures.

Whereas in DEC, kitchens and food were used as vehicles to get people to interact with each other, in Ciba-Geigy food, drink, and graciousness were handled very formally and carried additional symbolic meaning having to do with status and rank. Various senior officers of the company were pointed out to me, and I noticed that whenever anyone greeted another, it was always with their formal titles, usually Dr. This or Dr. That. Observable differences in deference and demeanor made it fairly easy to determine who was superior to whom within the organization. It was also obvious that the tables in the dining room were assigned to executives on the basis of status and that the hostess knew exactly the relative status of all her guests. I got to know a number of these executives over the years but another artifact was that I had to learn not to greet them in the dining room. If they acknowledged knowing me it would tell their peers that they needed consultation help, which apparently might be seen as a sign of weakness.

I reacted differently to the Ciba-Geigy and DEC environments. I liked the DEC environment more but could not decide whether it was the congruence with my U.S. identity, my experience in DEC that valued my more informal style, or the excitement of being in a start-up setting helping a company to grow versus trying to influence a very old culture to become more innovative. In performing a cultural analysis, a person's reactions are themselves artifacts of the culture that must be acknowledged and taken into account. It is undesirable to try to present any cultural analysis with total objectivity; not only would this be impossible, but a person's emotional reactions and biases are also primary data to be analyzed and understood.

I did not realize at the time that I was also confronting in these two companies an archetypal organizational issue: first, how to turn rampant creativity and innovation into a stable productive system and then, once a level of stability had been established, how to recapture some of the innovative capacity that is needed when a mature company faces changes in its technological, economic, and market environments. This issue has become a central focus of researchers on organizational structures and processes and has led to the concept of the “ambidextrous organization,” which is to be able to both maintain its “old” business while simultaneously creating and

protecting a new innovative business until it becomes mature and enables the company to survive in a new environment (O'Reilly & Tushman, 2016).

Espoused Beliefs and Values

Beliefs and values are usually elicited best when you ask about observed behavior or other artifacts that strike you as puzzling, anomalous, or inconsistent. If I asked managers in Ciba-Geigy why they always kept their doors closed, they would patiently and somewhat condescendingly explain to me that this was the only way they could get any work done and that they valued work very highly. Meetings were a necessary evil and were useful only for announcing decisions or gathering information. “Real work” was done by thinking things out, and that required quiet and concentration. In contrast, in DEC real work was accomplished by debating things out in meetings!

It was also pointed out to me that discussion among peers was not of great value and that important information would come from the boss or someone more technically expert. Formal and academic authority were highly respected, especially authority based on level of education and experience. The use of titles such as “Doctor” or “Professor” symbolized respect for the knowledge that education bestowed on people. Much of this had to do with a great respect for the science of chemistry and the contributions of laboratory research to product development. In Ciba-Geigy, as in DEC, a high value was placed on individual effort and contribution, but in Ciba-Geigy, no one ever went outside the chain of command and did things that would be out of line with what the boss had suggested.

In Ciba-Geigy, a high value was placed on product elegance and quality, and, as I discovered later, what might be called product significance. Ciba-Geigy managers felt very proud of the fact that their chemicals and drugs were useful for crop protection and creating the fertilizers that helped third-world countries deal with starvation, for curing diseases, and in other ways that helped to improve the world. The company had a clear worldwide identity that seemed to inform almost everything it did.

Assumptions—The Ciba-Geigy Company's Cultural Paradigm

Many of the values that were articulated gave a flavor to this company, but without digging deeper into assumptions, I could not fully understand how things worked. For example, the artifact that struck me most as I worked with this organization with the mandate to help it to become more innovative was the anomalous behavior connected with a memo I had written based on my learning how the company was successfully managing a difficult down-sizing process. I asked my contact client, Dr. Leupold, the director of management development, to distribute my memo to those managers he thought could most benefit from the information. Because he reported directly to Dr. Koechlin, he seemed like a natural conduit for communicating with those divisional, functional, and geographic managers who needed the information I was gathering. When I would return on a subsequent visit to the company and meet with one of the unit managers, without fail I would discover that he did not have the memo, but if he requested it from Dr. Leupold, it would be sent over almost immediately.

This pattern was puzzling and irritating, but its consistency clearly indicated that some strong underlying assumptions were at work here. When I asked one of my colleagues in the corporate staff unit that delivered training and other development programs to the organization why the information did not circulate freely, he revealed that he had similar problems in that he would develop a helpful intervention in one unit of the organization, but that other units would seek help *outside* the organization before they would “discover” that he had a solution that was better. The common denominator seemed to be that unsolicited ideas were generally not well received.

A third piece of information was that corporate marketing kept proposing integrated programs for all the divisions, only to be shot down with the comment: “How could there possibly be common training for Agri sales/marketing people who are slogging around in muddy fields talking to farmers, and well dressed MBAs visiting doctors in their hospital offices.”

My colleague and I had a long exploratory conversation about this observed behavior and jointly figured out what the explanation was. At Ciba-Geigy, when a manager was given a job, that job became the private

domain of that individual. Managers felt a strong sense of turf or ownership and made the assumption that each owner of a piece of the organization would be completely in charge and on top of his or her piece. Managers would be fully informed and make themselves experts in that area. Therefore, if someone provided some unsolicited information pertaining to the job, this was potentially an “invasion of privacy” and possibly an insult, as it implied that the manager did not already have this information or idea. The powerful metaphor that “giving someone unsolicited information was like walking into their home uninvited” came from a number of managers in subsequent interviews.

By not understanding this assumption, I had unwittingly put Dr. Leupold into the impossible position of risking insulting all his colleagues and peers if he had circulated my memos as I had asked. Interestingly enough, this kind of assumption means that even he could not articulate just why he had not followed my instructions. He was clearly uncomfortable and embarrassed about it but had no explanation until we uncovered the assumption about organizational turf and its symbolic meaning.

I realized that there was very little lateral communication occurring between units of the organization, so that new ideas developed in one unit never seemed to get outside that unit. If I inquired about cross-divisional meetings, for example, I would get blank stares and questions such as “Why would we do that?” Because the divisions were facing similar problems, it would obviously have been helpful to circulate some of the better ideas that came up in my interviews, supplemented with my own ideas based on my knowledge of what went on in other organizations. But here was a good example of a process that might work in the U.S. culture but might not even be considered in another macro culture. Of course, the irony in this example is that if I had understood this cultural characteristic I would have obtained a list of managers from Dr. Leupold and sent my memo directly to all of them. They would have accepted it as coming from the outside paid consultant and might even have viewed it as evidence that they were getting something useful from the outside expert.

Putting my interviews and direct observations together permitted me to construct a cultural paradigm for Ciba-Geigy, allowing, however, for the fact that I did not have nearly as much information in this instance as I had

in the DEC case. Because these assumptions were not as tightly linked as the DEC ones, I present them only as a list.

1. Scientific research is the source of truth and good ideas.
2. The mission is to make a better world through science and “important” products.
3. Truth and wisdom reside in those who have more education and experience.
4. The strength of the organization is in the expertness of each role occupant. A job is one’s own turf.
5. We are one family and take care of each other, but a family is a hierarchy and children have to obey.
6. There is enough time. Quality, accuracy, and truth are more important than speed.
7. Individual and organizational autonomy are the keys to success so long as one stays closely linked to one’s “parents.”

Analytical Comments. Ciba-Geigy had grown and achieved much of its success through fundamental discoveries made by a number of basic researchers in the company’s central research laboratories. Much of its basic culture could be attributed to the macro culture of chemistry, which is a formal hierarchic discipline in which experimentation has to be done carefully to avoid explosions, fires, and bad odors. DEC was based on electrical engineering in which “fooling around” was not only possible but often desirable.

Whereas in DEC truth was discovered through conflict and debate, in Ciba-Geigy truth had come more from the individual wisdom of the scientist or researcher. Both companies believed in the individual, but the differing assumptions about the nature of truth led to completely different attitudes toward authority and the role of conflict. In Ciba-Geigy, authority was much more respected, and conflict was to be avoided. The individual was given areas of freedom by the boss and then was totally respected in those areas. If role occupants were not sufficiently educated or skilled enough to make decisions, they were expected to train themselves. If they performed

poorly in the meantime, that would be tolerated for quite a while before a decision might be made to replace them.

In DEC if an individual was failing in a job the assumptions would be made that it was a mismatch rather than a personal failure and the individual would be allowed to negotiate a new assignment. In Ciba-Geigy, individuals would be expected to be good soldiers and do the job as best they could, and as long as they were perceived as doing their best they would be kept in the job. In DEC the individual was expected to negotiate his or her areas of freedom and then take full responsibility to report if things were not working out so that the job could be renegotiated, leading to a much more fluid job structure and much more vertical and lateral communication around work issues. Both companies had a “tenure” assumption that once people were accepted, they were likely to remain unless they failed in a major way or did something illegal or clearly immoral.

Both companies talked of being families, but the meaning of the word *family* was quite different in each company. In DEC, the essential assumption was that family members could fight, but they loved each other and could not lose membership. In Ciba-Geigy, the assumption was that parental authority should be respected and that children (employees and subordinate managers) should behave according to the rules and obey their parents. If they did so, they would be well treated, taken care of, and supported by the parents. They should not fight, and they should obey the rules and never be insubordinate.

Vertical and horizontal relationships in DEC were more personal whereas in Ciba-Geigy relationships were clearly more formal. This raises the interesting question of whether those differences reflected organizational culture and history or whether they reflected the macro cultures of the United States and Swiss Germany. If language is one of the major artifacts and characteristics of a national culture, one would note that English is a much less formal language than German, reflected even in the fact that in German the pronouns “du” and “sie” are used to differentiate how personal the relationship is meant to be.

In DEC, lifetime employment was implicit, whereas in Ciba-Geigy, it was taken for granted and informally affirmed. In each case, the family model reflected the wider macro-cultural assumptions of the countries in which these companies were located.

After I understood the Ciba-Geigy paradigm, I was able to figure out how to operate more effectively as a consultant. As I interviewed more managers and gathered information that would be relevant to what they were trying to do, instead of attempting to circulate memos to the various branches of the Ciba-Geigy organization through my contact client, Dr. Leupold, I found that if I gave information directly, even if it was unsolicited, it was accepted because I was an “expert.” If I wanted information to circulate, I sent it out to the relevant parties on my own initiative, or, if I thought it needed to circulate down into the organization, I gave it to the boss and attempted to convince him that the information would be relevant lower down.

If I really wanted to intervene by having managers do something different, I could accomplish this best by being an expert and formally recommending it to the CEO, Dr. Koechlin. If he liked the idea, he would then “order the troops to do it.” Inasmuch as he liked the career anchor idea, he ordered that everyone do the exercise for the summer program, but he mandated that in the following year all upper- and mid-level managers were to do the career anchor and job/role analysis, order their subordinates to do it, and then discuss it with them as part of the executive career development process. More will be said about Ciba-Geigy subsequently, but for now we need to explore the nesting concept again and ask a bigger question about macro cultures.

Can Organizational Cultures Be Stronger than National Cultures?

In terms of how cultures are nested within wider cultures, the question arises for both DEC and Ciba-Geigy as to what we would find in their various subsidiaries in other countries. They will, of course, vary for both type of organization and country of location, but some observations can be made about these two companies because they each had demonstrably strong organizational cultures.

I was able to visit subsidiaries of DEC in several European and Asian countries and found that at the artifactual level the DEC offices mirrored the headquarters in Maynard. The look of the place, the administrative procedures that were visible, and the informal climate looked and felt the

same. Clearly the DEC culture attempted to replicate itself in other countries, but the administrative and managerial employees were mostly local and spoke the local language, which led to some modifications of the company culture.

Such modifications were most noticeable in the product-design area, where, for example, the German customers wanted certain modifications in the product that led first to difficult negotiations with U.S. product managers and eventually to permitting local engineering staff to make the modifications for the local customers. The country managers were mostly local so that they could speak the language but were rotated periodically to headquarters assignments so that they could absorb the “essence” of the DEC culture.

The most extreme version of this kind of company indoctrination was the story of an HP plant manager whom I met in Singapore. He had been hired in Australia but before taking over the Singapore plant he was flown to California and spent two entire weeks “shadowing” CEO and founder David Packard to “absorb” the “HP Way.”

The strength of Ciba-Geigy’s culture was best illustrated by the previously mentioned example that one year when the U.S. subsidiary manager in New Jersey invited me to give a talk to the top layers of management on what I had learned about the headquarters culture in Basel. After I gave the talk, I got the shocked reaction, “My God, you have just described us.” Ciba-Geigy had evolved a systematic rotation of future executives into overseas assignments so that an effort to become more international would be reflected in all of its managers. The best example of the impact of that process was Samuel Koechlin himself, whose time in the United States subsidiary definitely influenced the Basel culture and made both the Basel and the U.S. cultures complex hybrids.

Summary and Conclusions

In the preceding two case analyses, I have tried to illustrate how organizational culture can be analyzed at several levels: (1) visible artifacts; (2) espoused beliefs, values, and behavioral norms; and (3) taken-for-granted basic underlying assumptions. Unless you dig down to the level of the assumptions, you cannot really decipher the artifacts, values, and norms. However, if you find some of those assumptions and explore their

interrelationship, you are really getting at the essence of the culture and can then explain a great deal of what goes on. This essence can sometimes be analyzed as a paradigm in that some organizations function by virtue of an interlocking, coordinated set of assumptions. Whereas each one alone might not make sense, the pattern explains the behavior and the success of the organization in overcoming its external and internal challenges.

We should not assume that even these paradigms describe the whole culture, nor should we assume that we would find the same paradigm operating in every part of the organization. How general the assumptions are throughout the organization should be investigated empirically depending on your purpose. As a researcher trying to describe a whole culture, your needs for completeness would be quite different than if you were an employee or customer dealing with a local unit. If you are a manager trying to change the culture or are considering a merger or acquisition, it is the essence, the DNA of the culture, that you would be most concerned about. I discovered these assumptions as a helper and consultant primarily through observation and exploring with inside informants some of the anomalies that I observed. It is when we do not understand something that we need to pursue vigorously why we do not, and the best way to search is to use our own ignorance and naïveté.

What are some the lessons to be learned from these cases, and what implications do they have for leadership? The most important lesson for me is the realization that culture is deep, pervasive, complex, patterned, and morally neutral. In both cases, I had to overcome my own cultural prejudices about the right and wrong way to do things, and to learn that culture simply exists. Both companies were successful in their respective technological, political, economic, and broader cultural environments for a long time, but both companies also experienced environmental changes that led to their disappearance as independent economic entities.

In both cases, the powerful influence of early leaders and historical circumstance was evident. Cultural assumptions have their roots in early group experience and in the pattern of success and failure experienced by these companies. Their current leaders strongly valued their cultures, were proud of them, and felt it important for members of their organizations to accept the assumptions. In both organizations, stories were told of misfits who left because they did not like the way the company operated, or who

were not hired in the first place because they either would have been disruptive or would not have liked it there anyway.

In both companies, leaders were struggling with changing environmental demands and were facing the issue of whether and how to evolve or change their ways of operating, but this was initially defined as reaffirmation of portions of the existing culture, not as changes in the culture. Though the companies were at different stages in their evolution, they both valued their cultures as important assets and were anxious to preserve and enhance them.

Finally, it is obvious that both companies reflected the national cultures in which they operated and the technologies that underlay their businesses. DEC was a U.S. company of creative electrical engineers evolving a brand new technology; Ciba-Geigy was a Swiss-German company of mostly highly educated chemical engineers working both with very old technologies (dye stuffs) and very new bio-chemical processes (pharmaceuticals). Electrical circuits and chemical processes require very different approaches and timetables for product development, which was pointed out to me many times. An important implication is that culture cannot really be understood without looking at core technologies, the occupations of organization members, and the macro-cultural context in which the organizations exist.

The major differences resulting from size, age, and leadership behavior were evident and will be spelled out more specifically in Chapter 11.

Questions for Readers

All readers should answer the following questions:

1. What struck you as most different in the two cases?
2. What do you think are the bases of those differences?
3. How much is attributable to their technologies?
4. How much is attributable to their national locations?
5. How much is attributable to their history, size, and age?
6. How well do you think you would fit into either company?

A DEVELOPMENTAL GOVERNMENT ORGANIZATION IN SINGAPORE

Can this culture model be usefully applied to a different type of organization? To test this I decided to include a shortened version of one of the chapters from my book on the culture study I had done in the early 1990s as a paid researcher in Singapore (Schein, 1996b).

Case 3: Singapore's Economic Development Board

In thirty years Singapore has gone from being a third world country with a per capita GDP of \$500 to having a per capita GDP of \$15,000 and being on the edge of the rich industrial world. No country has ever developed faster.

(Lester Thurow from Foreword to Schein, 1996b)

The case of Singapore illustrates the structure of cultural analysis very well, because the visible artifacts of the dictatorial repressive political regime that evolved there cannot be understood without locating the taken-for-granted basic assumptions the leaders had when they founded an independent Singapore in the early 1960s. Singapore's story begins with a vision shared by its political leader, Lee Kuan Yew, and his fellow British-educated colleagues, who were combining their shared vision with a desire to make this ex-British colony into a "global city with total business capabilities."

This shared vision can be thought of as the "espoused beliefs and values" of the culture model. What makes the case interesting is that it is one of the rare cases I have encountered in which the artifacts, the espoused values, and the underlying assumptions were well aligned with each other, so that one could easily see how the three levels were consistent with each other and could explain each other.

To implement that vision, Lee Kuan Yew and his colleagues decided in 1961 to create the Economic Development Board (EDB), a quasi-governmental agency to implement a plan to attract foreign investment. In a predominantly Chinese culture that is averse to failure, the EDB had to create an organization that would “avoid punishing those who fail in the course of testing the limits of the system, but instead to punish those who are incompetent and those who do not learn from failure. Identify failure; change what does not work; create a learning environment. Easy advice—but hard to do unless one evolves a culture that supports such attitudes . . . create a long-range vision, build a team, draw out the best in team members. Demand total loyalty to the mission and a 120 percent commitment from everyone. Provide one-stop shopping for the clients from a totally professional organization devoted to teamwork, open communications, and a borderless organization. The rules are clear, there is no corruption, and integrity is total” (Thurow, Foreword to Schein, 1996b).

The EDB was very successful and decided in 1990 to have someone document its story. The EDB leaders originally hired a journalist to write this story but decided that it was their *culture* that was the key to their success, so they looked for someone who knew about culture. They consulted Lester Thurow, who was then Dean of the MIT Sloan School where I taught. He suggested that they approach me, which led to my agreeing to investigate their success story from three perspectives: (1) their view of themselves, (2) the view of the various CEOs who had decided to invest in Singapore by building plants and research organizations there, and (3) my analysis of the artifacts, espoused values, and basic assumptions that could be inferred from all of these data.

The EDB’s view of itself was obtained through intensive interviews of all the leaders who had created and supported the EDB through the past three decades during several two-week visits to Singapore in 1994 and 1995. I then located and interviewed as many CEOs or other senior executives who had made the decision to invest in Singapore to determine why they had done so and how it had worked out. My analysis was supplemented by direct observation of how the EDB worked along with my attendance at various group meetings that had been set up to provide me with more information. The leaders of the EDB were clearly very proud of their accomplishments and wanted the study to document the positive elements

of what they had done, but they also made it very clear that they wanted to learn from my analysis what their weak spots and future learning challenges might be. In other words, I was to be critical as well as positive.

The entire complex story of thirty years of development is told in my book *Strategic Pragmatism: The Culture of Singapore's Economic Development Board* (1996b) and is abstracted in this chapter.

The EDB Nested Cultural Paradigms

The structural model of artifacts, espoused values, and basic underlying assumptions proved to be necessary to make sense of all the interviews and observational information that I had gathered over the period of a year or so. The concept of cultures nested within other cultures was immediately evident in that the way the EDB setup reflected both the Chinese origins of the leaders and the impact of their British education and colonial experience. Furthermore, Singapore was originally embedded in the Malaysian Federation, which created cross-cultural tensions at the national level. Then, after Singapore broke away and became independent in 1965, that led to a period of economic *interdependence* because Singapore had no water supply of its own.

The basic assumptions that could be inferred and tested with the “insiders” thus fell into a “contextual” set that reflected the nesting and an “organizational” set that reflected the way in which the EDB as a separate organization managed its external and internal relationships. The contextual paradigm consisted primarily of a set of assumptions that Singapore’s leaders held about economic development. These assumptions were shared by the EDB, but they also provided a broader context within which the EDB operated. The organizational paradigm consisted of a set of assumptions about how the EDB structured and managed itself.

1. The Contextual Paradigm: Assumptions about the Role of Government in Economic Development

The contextual paradigm consists of six interlocking and interrelated shared basic assumptions that reflect the mental models of the early leaders of Singapore and are largely taken for granted today. These assumptions are

shared by Singapore's government in general and thus provide a cultural context within which the EDB operates. At the same time they are assumptions held by leaders and members of the EDB itself and thus influence more directly how the EDB operates. These assumptions led to the creation of the EDB and provided the espoused values that influenced how the EDB would define its mission and organize itself. We have here a case where the espoused beliefs and values are congruent with the observed artifacts—the work of the EDB.

1a. “State Capitalism.” *Singapore's leaders and the EDB assumed and took for granted that government could and should play an active entrepreneurial role in economic development and should therefore exercise leadership through a quasi-governmental statutory board such as the EDB.*

1b. Absolute Long-Range Political Stability. A closely connected second core assumption that came to dominate Singaporean thinking and action is really a cluster of three interconnected assumptions that can be stated as follows. *Singapore's political leaders assumed (1) that economic development must precede political development, (2) that long-range successful economic development could occur only if there was political stability, and (3) that political stability could be achieved and maintained only by firm but benign government controls that steer all segments of the society.*

This was, of course, the most critical assumption to understand because it was the basis of the observed dictatorial regime that was set up around political and civil behavior. For example, severe punishment for littering or urinating in an elevator was justified on the grounds that Western businessmen would feel more confident working in a squeaky clean city. The rules and the heavy punishment were visible artifacts, but few observers really understood that there was a deep economic development assumption behind the rules and their implementation. The basic assumption that economic development justifies social control was never challenged as long as Singapore was succeeding in its development efforts.

1c. Collaboration among Sectors. *Singapore's political leaders assumed that economic development could succeed only if business, labor, and government actively collaborated with each other in fulfilling the common goal of building the nation (“Singapore, Inc.”).*

Interorganizational collaboration was considered vital to provide the incentives and infrastructure needed to develop the manufacturing and service sectors—that is, roads, communication facilities, land, financial support for investment and training, a well-trained and motivated labor pool, housing, and so forth. One of the most notable aspects of this way of thinking was the decision to give the trade unions some responsibilities as owners and managers by having them own and operate one of the taxi companies and one of the insurance companies in Singapore. Some industrial-relations analysts might regard this as co-optation and undermining of the labor movement, but from Singapore's rulers' point of view, getting labor on their side was paramount, especially in view of the proximity of a communist China to the north.

1d. An Incorruptible, Competent Civil Service. *Singapore's political leaders assumed that favorable economic conditions for investors would be guaranteed only if the government and civil service were competent and incorruptible and operated with an open and consistent set of rules that were vigorously enforced.*

Here again we see nesting in that this assumption reflects the Chinese cultural legacy that rulers must be exemplars of virtue, the British traditions of a “clean civil service,” and the early recognition on the part of Singapore's leaders that overseas investors would be attracted only to a developing nation in which there was not only political stability but also a competent government, a clear set of rules, and an absence of corruption.

1e. Primacy of People and Meritocracy. *Singapore's leaders assumed that the only resource they had were its people and their potential; it must, therefore, pick the best of them and develop them.*

Out of these assumptions grew the political decision to provide everyone with jobs and housing, to mandate English as the official language, and to create a lucrative government-sponsored scholarship program that would send the best students overseas to the best universities in exchange for a number of years of government service at salaries that were competitive with those offered by private industry.

1f. Strategic Pragmatism. Taken together these five assumptions describe what might best be labeled “strategic pragmatism” in that there was a

clear long-range strategy reflecting Chinese cultural biases, but the implementation was thought through on a pragmatic daily basis and expressed in very detailed rules of how life was to be lived. This attention to pragmatic detail seemed more Western and was, in any case, aimed to attract Western business.

Singapore's leaders assumed that the survival of the city-state required a very long-range plan but that the implementation of that plan had to begin immediately on a very practical level with the creation of the EDB.

These long-range aspects included the decision to stabilize Singapore by attracting businesses that would make major capital investments and, therefore, would be committed to remaining. The leaders were very aware of how vulnerable city-states are that depend only on their port and shipping.

2. The Cultural Paradigm of the EDB as an Organization

The culture of the EDB as a separate organization nested within the contextual paradigm is a set of paradoxes and anomalies from a Western point of view, but its basic assumptions are consistent with each other and enabled the organization to function effectively. This paradigm is best described in terms of six basic assumptions that came to dominate daily activities and the way the EDB organized itself.

2a. Teamwork: Individualistic Groupism. *The EDB assumed that the best kind of leadership is to build a team and that the ultimate responsibility of team members is to contribute to the maximum of their ability.*

The EDB employees seemed equally comfortable working competitively with each other in producing results while being completely dedicated to the team, the EDB, and the nation-state. One sees here a cultural legacy from Confucian principles of concern for family combined with a Western concept of individual achievement. Underlying this assumption was the reality that the EDB always had to function as a team because it was a small organization in which members therefore had to help each other. However, this pragmatic reason for teamwork was also supported by a cultural tendency to be comfortable in a team setting, a kind of comfort that is noticeably absent in many Western teams. At the same time, all of the officers were educated in settings where individual achievement was highly valued,

and they were constantly exposed to multinational company managers who lived by individualistic competitive rules and were encouraged to develop their individual careers within the EDB.

The organization attracted very strong individualists who did, in fact, compete with each other and noticed it when someone was promoted ahead of them. The competition was mitigated by the fact that they were all kept so busy that they had very little time to worry about each other's accomplishments. If you were too individualistic or political, you very quickly lost credibility with your colleagues and found yourself unable to do very much. So the two major distortions that could occur in the communications system—sitting on information or, at the other extreme, hyping up information and exaggerating what you know—were both mitigated by the need to maintain credibility and trustworthiness. To get anything done you had to build support, and to build support you had to maintain your credibility. Everything in the EDB had to be highly coordinated.

To be a successful performer in this kind of high-pressure family-team required a complex ability to collaborate with others and be a true team player while, at the same time, exposing individual talents and skills for purposes of promotion and career progress. This kind of balancing act was facilitated by the ability to think clearly, articulate clearly, write clearly, and be able to convince others to “join one's team” in support of a project. In other words, individual talent showed up most in the individual's quality of thinking and communicating, which was then tested in one's ability to create and work in a team. Individual achievement was recognized with awards and other forms of recognition.

The EDB's view of itself as a team and practically a family was partly based on everyone knowing everyone else. This level of acquaintance was maintained through many informal activities such as the weekly Friday afternoon teas; company functions such as picnics and sports outings that families were encouraged to attend; the monthly newsletter entitled “Network,” in which a variety of personal news items, especially awards and individual accomplishments were given publicity; and the encouragement of romantic attachments among employees symbolized by the great pride in the number of married couples who had met each other as EDB employees and remained in their jobs.

Also supportive of this team spirit was a set of very flexible personnel policies that allowed part-time assignments or jobs in which one would not be required to travel if family responsibilities made travel difficult. The work of the EDB was so intrinsically motivating that there was never any question of loss of work motivation. Rather, the EDB tried to accommodate the needs of each employee because each was viewed as being valuable.

2b. Cosmopolitan Technocracy. If Singapore's fate rested on its ability to attract overseas investors, the EDB had to be able to deal with many other cultures and their officers had to be what sociologists have called "cosmopolitan" in their orientation. At the same time, to bring in the right investors and provide good services for them once they were in Singapore, the EDB had to staff itself with a cadre of people who would be both competent marketers or salesmen and entrepreneurs.

The EDB leaders and managers, therefore, had to be comfortable and knowledgeable in the global multicultural arena and at the same time very much in touch with the situation in Singapore and the potential of synergy between the multinationals and local industry. Their cartoon version of themselves depicted them as supermen and superwomen. To make all of this work required a particular personnel philosophy that could best be stated in terms of the following basic assumptions, many of which were explicitly espoused. *The EDB assumed that it could only succeed if it recruited*

1. *The "best and the brightest" based on scholastic performance.*
2. *Officers with a "cosmopolitan orientation" based on overseas education and interest in working with and in overseas business settings.*
3. *Officers who were technically oriented and trained because the kind of businesses that were to be promoted were usually technically based.*
4. *Officers who had high levels of personal initiative that enabled them to work in unpredictable and uncharted business and government arenas.*
5. *Officers and managers who were team oriented and had high levels of interpersonal skill to deal with multiple cultures, with multiple hierarchical levels, and across organizational boundaries of all kinds.*

Many of the qualities that were sought in the “officers” were already present in the leaders who created the EDB because they had grown up in a multicultural environment themselves—British, Malay, Tamil, and Chinese. Many of them had technical backgrounds and biased the education toward engineering and science from the outset, once again reminding us that the culture of the EDB was nested not only in the Singaporean culture but in the many other cultures that the early leaders brought into the picture.

2c. Boundaryless Organization: Modulated Openness. The EDB emphasized timely, accurate, and widely dispersed information as essential to decision making, often describing itself as a “boundaryless” organization. Two basic assumptions operated behind this principle, one referring to internal operations and one linking back to the contextual assumption about sector collaboration.

The EDB assumed that the only way it could fulfill its function effectively was for all managers, officers, and other relevant employees of the organization to be fully informed about all projects at all times. The EDB assumed that the only way it could fulfill its function was to develop and maintain open channels to the other sectors of the government as well as to private and labor sectors.

For the EDB to make quick and valid decisions about investments and investors, it believed that it was necessary for all relevant information for any given project to be available to all members of the organizations who might have an input to the decision, and certainly to the higher-level decision makers. This assumption resulted in the extensive global communication system that the EDB set up; its willingness to spend money on communications, travel, and meetings; and a standardized reporting system that allowed information to be efficiently centralized. Everything had to be written down, training was provided to employees in written communication, and, perhaps most important of all, the norm was articulated that “one must pass on all relevant information truthfully and not use information as a personal source of control or power.”

The “modulated openness” referred to the potential problems of simultaneously working with many investor clients, many of whom were competing with each other. For example, Hewlett-Packard and Digital Equipment Corp. were both considering setting up manufacturing plants in Singapore.

If highly confidential plans were revealed, it was not always clear even within the EDB whose interests could be hurt by too much exposure of such plans. EDB officers therefore had to be very situationally careful while espousing maximum openness.

2d. Non-Hierarchical Hierarchy: The Boss as Patron, Coach, and Colleague. The EDB culture implicitly assumed that managers could succeed only if they had a strong sense of autonomy in performing their task, a willingness to initiate decisions through formal proposals up the hierarchy, a willingness to be open and frank in revealing information up the hierarchy, a willingness to go around the hierarchy when tasks required it, and the ability to work with higher levels of management in the client organizations.

At the same time they assumed that managers had to show suitable deference to superiors when appropriate (particularly in public), to seek and accept guidance from above in revising proposals and in making decisions, to show good judgment in keeping their superiors fully informed when going around the hierarchy, and to show appropriate humility when being coached and guided by superiors and when dealing with higher-ranking managers in client companies.

The best way to characterize this set of relationships is to note that EDBers were expected to perform as one would in a boundaryless Western organization in which hierarchy is downplayed while performing as one would in an Asian (Chinese) organization in which deference and hierarchy are dominant. What the young senior officer had to learn in entering this organization was how to do that—how to develop the judgment and interpersonal skills to perform according to both sets of norms.

Interpersonal skills were especially relevant because, on the one hand, boundaries were nonexistent in the sense that one could always walk into a senior executive's office and talk very frankly to him. On the other hand, EDB officers knew that department heads might feel at a disadvantage when their subordinates went around them. Consequently, one of the important interpersonal skills was knowing how to keep the department heads feeling sufficiently secure so that they would not feel threatened if either a subordinate or a superior went around them. The implication is that one of the most important aspects of being socialized into the EDB was to learn

the rules and develop the skills of being open and nonhierarchical without threatening the hierarchy.

2e. Extended Trust Relationships: Clients as Partners and Friends. One of the important distinguishing features of the EDB was its conception that the overseas investor was to become a friend and partner and that the relationship was to be a long-term one that would be of mutual benefit to the company and to Singapore. Implicit in this concept was not only the long-range strategic goal but an extension of the Chinese philosophy of *guanxi*, or building trusted connections that can be used in the future. Whereas in the old Chinese system such connections were limited by personal acquaintance and patterns of mutual obligation that extended out from the family and clan, the EDB concept was a much more Western notion of forging strategic alliances and partnerships with investing companies to create the kind of industrial system that the strategy envisioned.

The EDB had set aside considerable investment funds to make such partnerships real by allowing itself to be an equity partner as well; the purpose was not to invest to make more money, but to ensure that the enterprise would succeed. Once a business was a going concern, the EDB always intended to sell off its share to make the money available for the next project. This general philosophy rested on two basic assumptions.

The EDB assumed that it could succeed only if it fully understood the needs of its clients (potential and present investors) and collaborated with them in solving their problems efficiently but without compromising its own basic goals, plans, or rules (strategic pragmatism). The EDB assumed that Singapore's long-range mission could be fulfilled only if initial investors continued to invest and became committed to transferring technology and training to Singapore's labor force. Such continued investment could be achieved only if the EDB became friends and partners with its initial investors.

It was never enough just to bring the investors in. Once they were in, they inevitably developed new needs and problems, and it was the EDB officer whom they would call when they needed help. Such help often laid the groundwork for further investment and an expanded relationship with Singapore. In that regard, possibly one of the most important dimensions of the EDB and Singapore culture was the attitude toward time. On the one hand, there was a lot of emphasis on long-range planning and figuring out

how to create a set of incentives and activities that would encourage the investor, who was thinking about the long haul. On the other hand, there was tremendous pride in being a good host and doing whatever it took to help a foreign investor succeed in the short run. EDBers thought of themselves as instant problem solvers.

At the artifactual level, the long-range point of view was evident in the willingness to spend large amounts of money on training and education. The educational establishment was charged with offering the kinds of curricula that would fit the long-range needs of the country. Evidence for the short-run pragmatism comes from the frequent changes in social policy that the government was willing to undertake when a given policy did not accomplish what was intended. Singapore as a state and the EDB as an organization both displayed the ability to change course rapidly if their analysis indicated a need.

Singapore was neither a long-range planner like Japan nor a short-range pragmatist like Hong Kong or many Western countries that are driven by a monthly or quarterly business model. It is both, and it manages somehow to combine the two by having clear long-range goals and visions that are widely circulated throughout the nation-state and, at the same time, a sense that those goals and visions cannot be achieved if the daily problems of helping the industrial establishment are not solved immediately. The key to this combination is the building of relationships with investors so that their long-range interests will coincide with Singapore's. The day-to-day solving of problems ensures the solidity of the long-range partnership and friendship.

2f. Commitment to Learning and Innovation. Just as “strategic pragmatism” served as a kind of integrative assumption around the contextual paradigm, the commitment to learning and innovation serves to tie together the assumptions of the organizational paradigm. In a sense, commitment to innovation is also paradoxical because so much cultural analysis of Asian societies emphasizes a fatalistic view, acceptance of harmony with nature, and commitment to stability and harmony within the social structure. Clearly Singapore has blended whatever Asian legacy it has with a more Western proactive stance that anything is possible, symbolized in Singapore by the often-heard phrase “dare to dream.” In assumption form this can be stated as follows:

The EDB (and the government of Singapore) assumed that the only way it could fulfill its vision of development was to learn from others and its own experience, and to continuously innovate in dealing with whatever problems were discovered to stand in the way of achieving the vision.

This attitude goes back to the early leaders' willingness to learn from other countries and from various non-Singaporean advisers; it was most clearly demonstrated in the continuous changing and refining of social policy. It is true that the policies are viewed by many as excessively controlling and a real restriction on the freedom of the individual, but in that perception one may miss the equally important point that the policies are constantly changing in response to new data. One of the important roles of the EDB is to stay in touch with what is going on in the rest of the world, which they did through a vast network of offices in all the major industrial centers. The knowledge acquired in this network became a major source of feedback to the government and provided the data needed to adjust government policy.

Various corporate seminars on marketing and strategic planning put on by the EDB became visible manifestations of the desire to stay open and learn. The EDB openly embraced the concept of the "learning organization," using Peter Senge's *The Fifth Discipline* and concepts of systems thinking (Senge, 1990).

Summary and Conclusions: The Multiple Implications of the Three Cases

The reader may well wonder why bother with such detailed cases. Should we not be looking for broad generalizations about organizational and national culture? There are several reasons for studying detailed cases.

First, the devil is in the details. Humans are complex at the personality level; groups, organizations, and nations are complex at the cultural level. We will later review some typologies that are intended to provide simpler models into which to sort cultures. For example, one popular model is to think of organizations as "markets," "hierarchies," or "clans" (Ouchi, 1981; Williamson, 1975). By that classification DEC, Ciba-Geigy, and Singapore's EDB would all have to be called clans, and we can see immediately that this would lose some of the important ways in which the clannish family feeling

in each organization played out differently. The three organizations were at different stages of development, which strongly influenced the way the cultures evolved, and they were nested in entirely different national cultures.

Second, cultural details have to be understood to determine how these organizations evolved. The DNA of DEC's culture survived while the organization as an economic entity failed. Ciba-Geigy changed some of its externally focused DNA in abandoning some chemical businesses while enhancing the pharmaceutical business, but held on strongly to its way of dealing with people in making the changes; the EDB continued to be successful in helping Singapore to grow as a viable economic and politically stable city-state, thus reinforcing the complex mix of Asian and Western values that its cultural paradigm illustrates.

Third, how things work inside a culture and how employees and managers might feel on a daily basis can be inferred only by understanding the interrelationship of the cultural components, what I have labeled the cultural "paradigm" in each organization. As each case illustrates, to understand the paradigm one must identify both how the cultural components interact with each other and how they interact with the components of the cultures in which they are nested. DEC's sense of indifference to cost reduction was very much related to its individualistic values of not being willing to fire "good people." Ciba-Geigy's carefully orchestrated downsizing reflected to a considerable degree the Swiss-German and Basel community values. The EDB learned how to combine the traditional Chinese values with Western values to create its successes.

Fourth, when we examine the dynamics of *managed* culture evolution and change, we will see that the strategy and tactics of intervening successfully require a more detailed knowledge of the cultural elements and how they interact. We will not need analyses at the level of complexity of these cases, but we will need a process for quickly identifying which cultural elements will help us manage the desired changes and which ones will hinder us and become targets of change.

We have now reviewed the defining structure of culture and illustrated it with several detailed case studies. The cultural *dynamics* were covered to some degree in the cases, and the way organizational cultures are nested in macro cultures was illustrated. We now need to understand more about how to think about and assess those macro cultures.

Questions for Readers

Ask yourself the following questions:

1. In what way is this “government” agency different from a business?
2. How do the national cultures affect how the EDB functions?
3. What is the impact on the culture of what the organization’s task is?

